

Pathogenic Bacteria and Key Diseases

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Learning Objectives

- q Appreciate the concept of Normal Flora of the human body
- q Explain Key mechanisms of disease
- q Link organisms to common clinical syndromes
- q Outline basic diagnostic and treatment principles

Normal Flora

- **Normal flora** organisms that inhabit the body surfaces exposed to the environment, such as the skin, oropharynx, intestinal tract, and vagina.
- Members of the normal flora differ in number and kind at various anatomic sites
- Members of the normal flora are **low-virulence** organisms.
- They are nonpathogenic in their usual anatomic sites
- Generally, viruses and parasites are not recognized as part of normal flora of the human body

Members of Human Normal Flora

- **Skin**-the predominant member of the normal flora of the skin is *Staphylococcus epidermidis*. It is an important cause of infections of prosthetic heart valves and prosthetic joints.
 - ✓ *Staphylococcus aureus* is also present on the skin, but the **nose** is its site. It causes abscesses in the skin and in many other organs
- **Oropharynx**. The main members of the normal flora of the mouth and throat are the viridans streptococci.
 - ✓ Viridans streptococci are the most common cause of subacute endocarditis.

Members of Human Normal Flora

- **Gastro-intestinal tract** - The two most important members of the colonic flora that cause disease are the anaerobe *Bacteroides fragilis* and the facultative *Escherichia coli*.
- **Vagina** - Lactobacilli are the predominant normal flora organisms in the vagina.
 - ✓ They keep the pH of the vagina low, which inhibits the growth of organisms such as *Candida albicans*, an important cause of vaginitis

Members of Human Normal Flora

- **Urethra** - the outer third of the urethra contains a mixture of bacteria, primarily *S. epidermidis*.
 - ✓ The female urethra can become colonized with fecal flora such as *E. coli*, which predisposes to urinary tract infections.

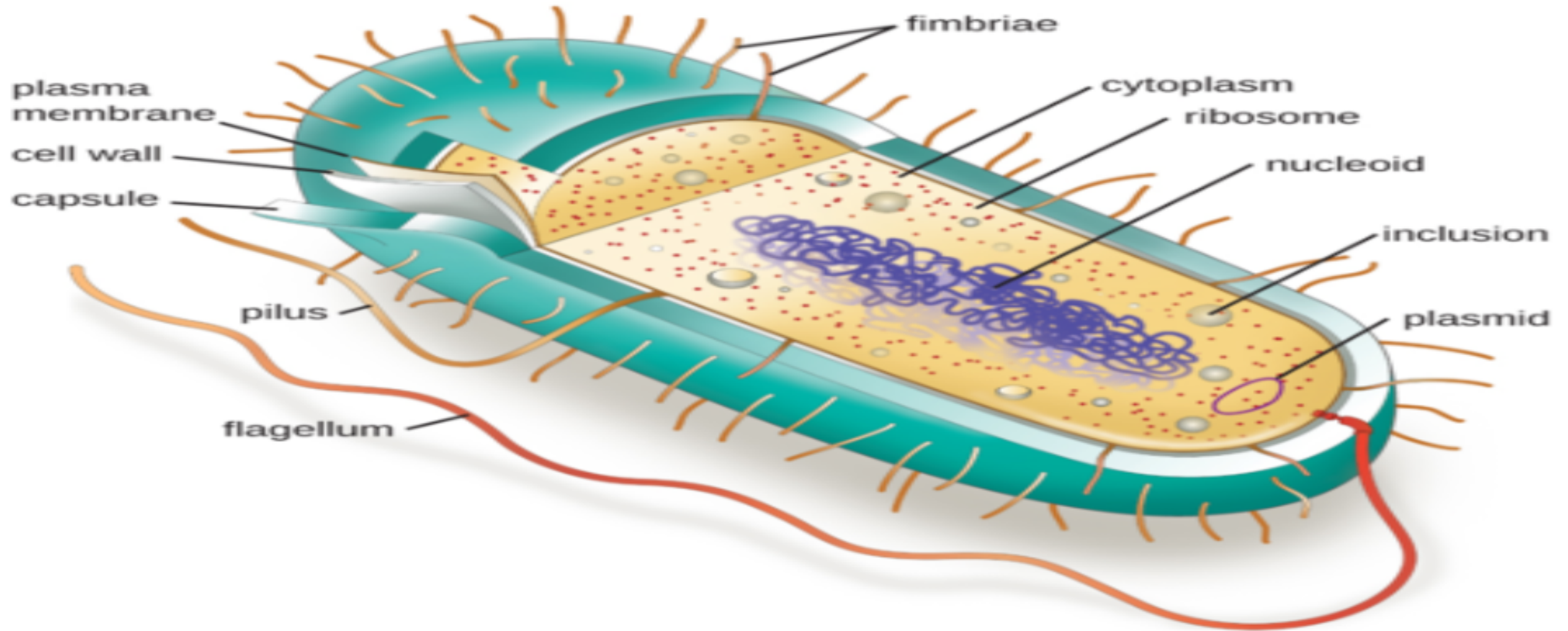
Role of the Normal Flora

- They can cause disease particularly in immunocompromised and debilitated individuals.
- They constitute a protective host defense mechanism.
 - ✓ The nonpathogenic resident bacteria occupy attachment sites on the skin and mucosa that can interfere with colonization by pathogenic bacteria. The ability of members of the normal flora to limit the growth of pathogens is called **colonization resistance**.
 - ✓ If the normal flora is suppressed, pathogens may grow and cause disease.
- They may serve a nutritional function.
 - ✓ The intestinal bacteria produce several B vitamins and vitamin K.

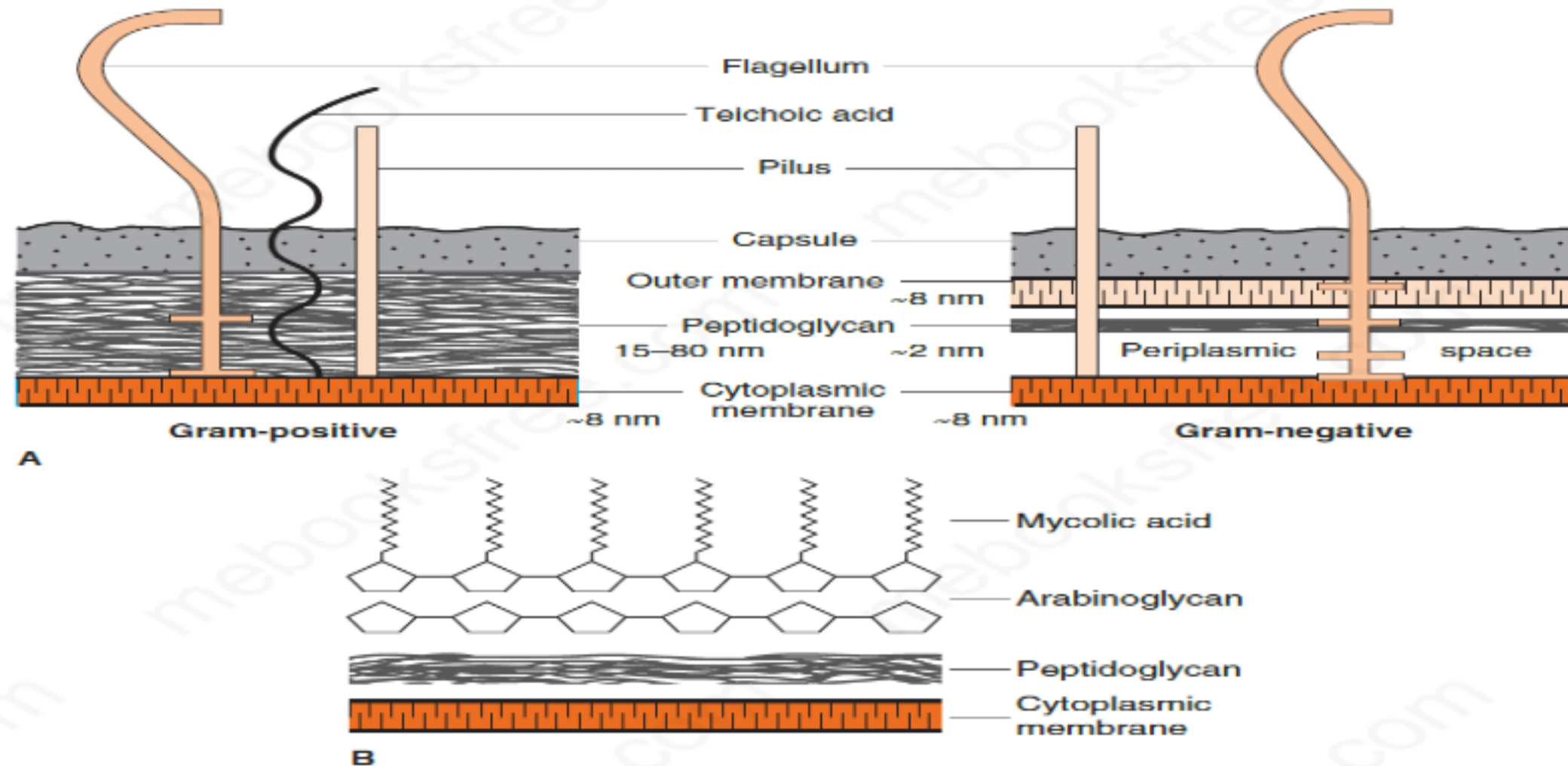
When do normal human flora cause disease

- Barrier breakdown (*Staph aureus* causing wound infections)
- Immunosuppression (HIV, chemotherapy, steroids, malnutrition, extremes of age)
- Disruption of normal flora (broad spectrum antibiotics, overgrowth of resistant or opportunistic organisms eg *Clostridioides difficile*)
- Translocation to abnormal or unusual sites (eg *E coli* causing UTI)
- Overgrowth due to favourable conditions (pH changes, hormones, nutrients) eg vaginal candidiasis after antibiotics or pregnancy
- Foreign bodies encouraging biofilm formations (*Staph.*

Structure of typical bacterial pathogen



Structure of typical bacterial pathogen cell wall



Structure of typical bacterial pathogen

Structure	Chemical Composition	Function
Essential components		
Cell wall		
Peptidoglycan	Glycan (sugar) backbone with peptide side chains that are cross-linked	Gives rigid support, protects against osmotic pressure, is the site of action of penicillins and cephalosporins, and is degraded by lysozyme
Outer membrane of gram-negative bacteria	Lipid A	Toxic component of endotoxin
Surface fibers of gram-positive bacteria	Polysaccharide Teichoic acid	Major surface antigen used frequently in laboratory diagnosis Major surface antigen but rarely used in laboratory diagnosis
Plasma membrane	Lipoprotein bilayer without sterols	Site of oxidative and transport enzymes
Ribosome	RNA and protein in 50S and 30S subunits	Protein synthesis; site of action of aminoglycosides, erythromycin, tetracyclines, and chloramphenicol
Nucleoid	DNA	Genetic material
Mesosome	Invagination of plasma membrane	Participates in cell division and secretion
Periplasm	Space between plasma membrane and outer membrane	Contains many hydrolytic enzymes, including β -lactamases
Nonessential components		
Capsule	Polysaccharide ¹	Protects against phagocytosis
Pilus or fimbria	Glycoprotein	Two types: (1) mediates attachment to cell surfaces; (2) sex pilus mediates attachment of two bacteria during conjugation
Flagellum	Protein	Motility
Spore	Keratinlike coat, dipicolinic acid	Provides resistance to dehydration, heat, and chemicals
Plasmid	DNA	Contains a variety of genes for antibiotic resistance and toxins
Granule	Glycogen, lipids, polyphosphates	Site of nutrients in cytoplasm
Glycocalyx	Polysaccharide	Mediates adherence to surfaces

Bacterial Disease Pathogenesis

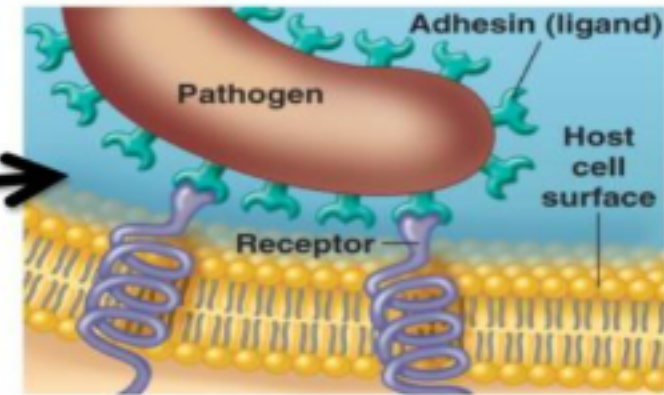
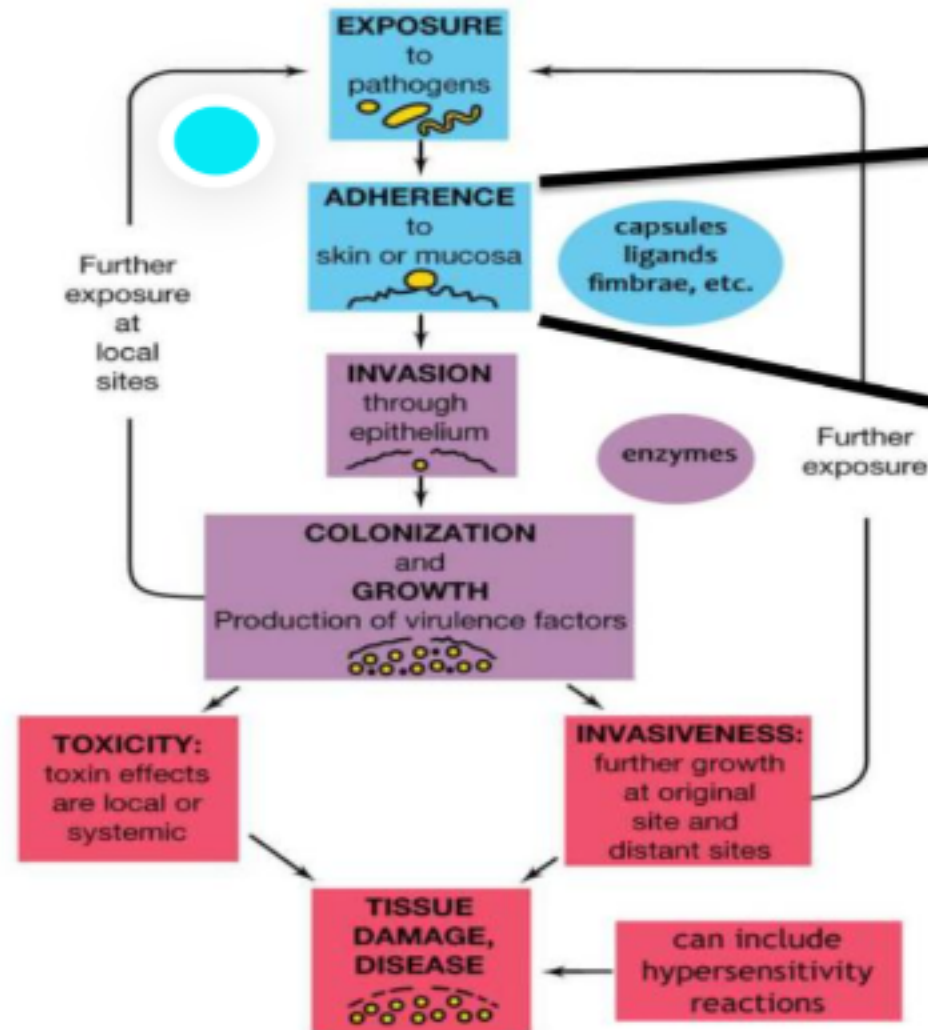
- Pathogenesis is the process by which a disease or disorder develops
- Infectious disease is established when microorganisms over-power the host defences
- There is usually a complex inter-play of factors between the host and the infecting microorganism before disease is established
- From the Infecting organism's perspective, the two critical determinants in overpowering the host are
 - the **number of organisms** to which the host, or person, is exposed and
 - the **virulence** of these organisms.

Stages of Bacterial Pathogenesis

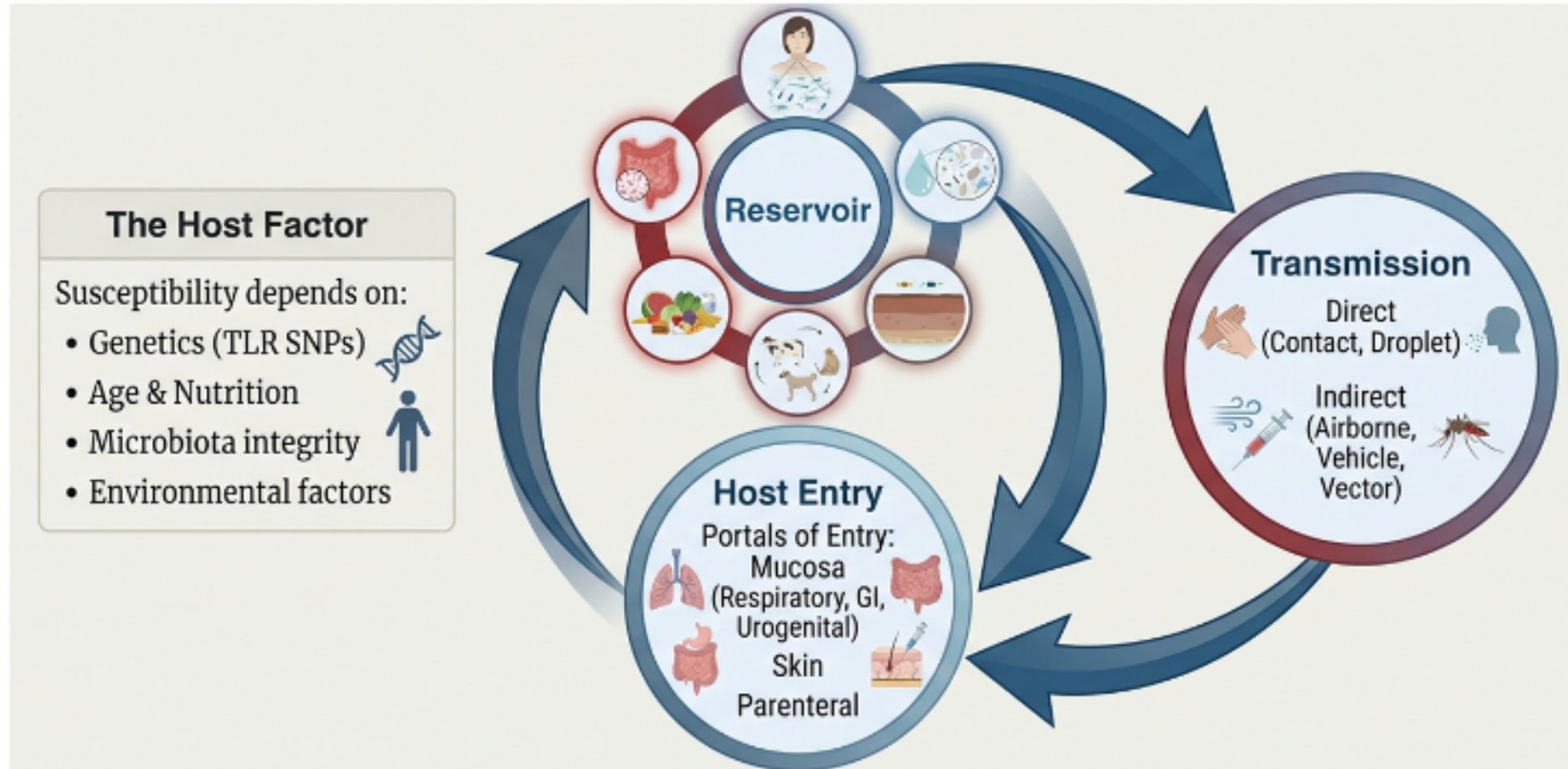
- Bacterial infection could be acquired from an external source or from the normal flora of the host
- A generalized sequence of the stages of infection is as follows:
 1. Transmission into the portal of entry.
 2. Adherence to mucous membranes.
 3. Evasion of primary host defences such as skin or stomach acid.
 4. Colonization – growth of the bacteria at the site of adherence.
 5. Disease symptoms caused by toxin production or invasion or immunopathogenesis accompanied by inflammation.
 6. Host responses, both nonspecific and specific immunity
 7. Progression or resolution of the disease

Bacterial Disease Pathogenesis

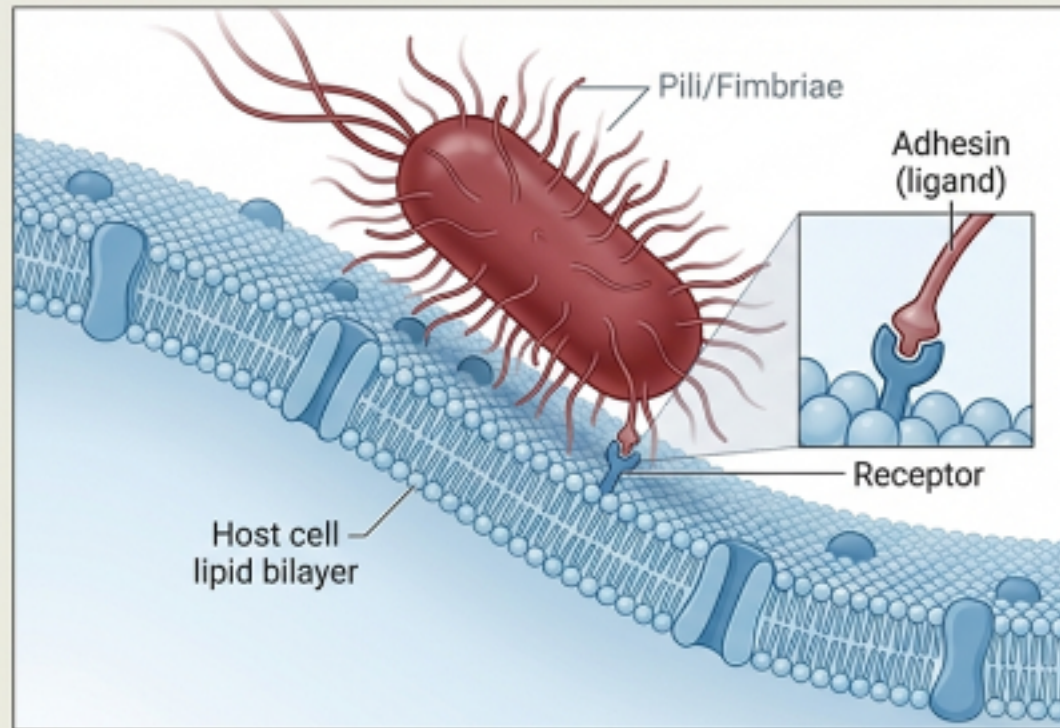
pathogenesis



Stages of Bacterial pathogenesis – Exposure



Stages of Bacterial pathogenesis – Adherence



- Adherence Tools:
 - **Pili/Fimbriae:** *N. gonorrhoeae* & *E. coli* (Urinary tract attachment).
 - **Glycocalyx/Slime:** *S. epidermidis* (Heart valves/Catheters).
 - **Curli Proteins:** *E. coli* & *Salmonella* (Bind endothelium).
 - **Capsules:** Sticky surface + Anti-phagocytosis.

Principle: Adherence is essential. Mutants without these mechanisms are non-pathogenic.

- These structures are collectively called adhesins

Stages of Bacterial pathogenesis – Invasion and Toxicity

Phase III: Invasion & Toxicity



Enzymatic Invasion (The Drill)

- **Collagenase/Hyaluronidase:** Degrades tissue “cement” (*S. pyogenes*).
- **Coagulase:** Forms fibrin walls to block phagocytes (*S. aureus*).
- **IgA Protease:** Destroys mucosal antibodies (*N. gonorrhoeae*).



Exotoxins (The Sniper)

Secreted proteins. Highly toxic and specific.

Examples: Neurotoxins (Botulism), Enterotoxins (Cholera).



Endotoxins (The Bomb)

Lipid A (LPS) from Gram-negative walls. Released on cell lysis.

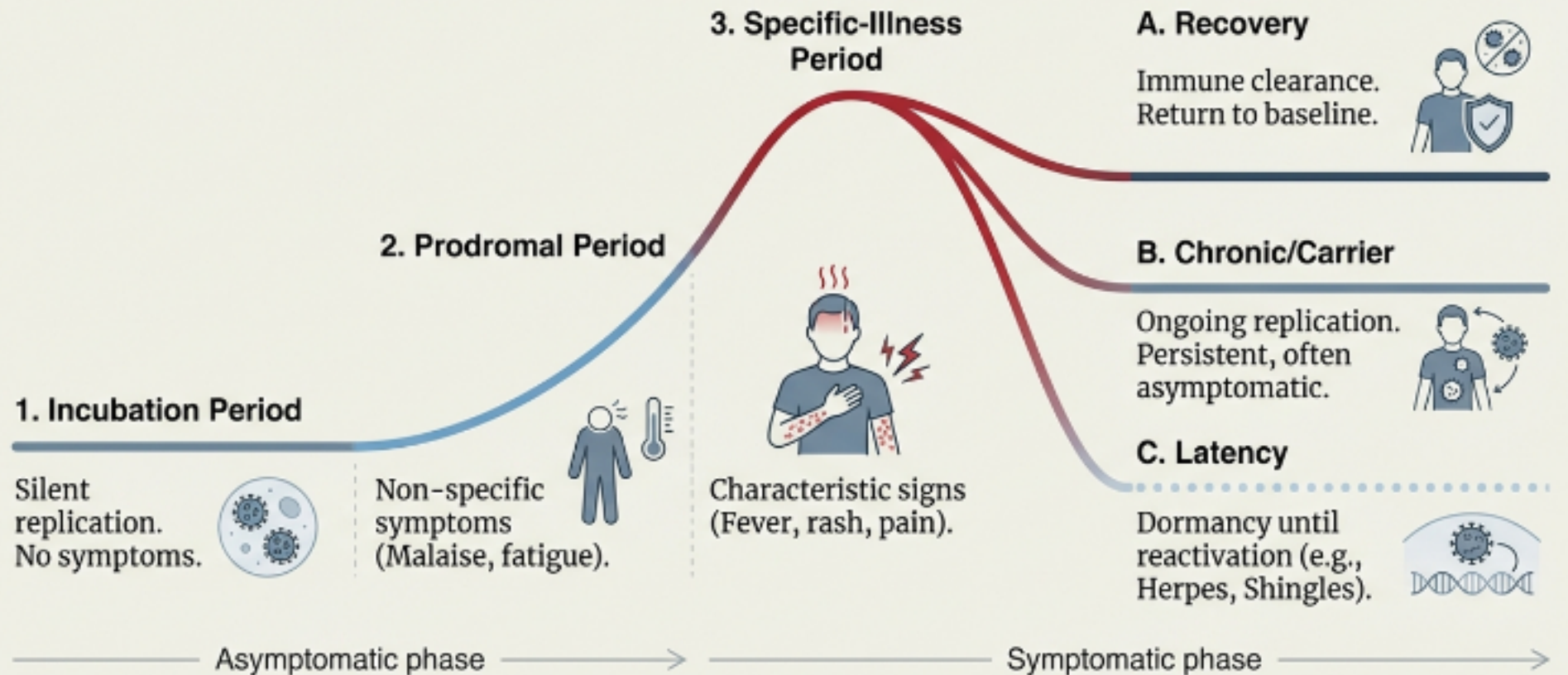
Effect: Generalized fever, inflammation, septic shock.

Bacterial Pathogenesis - Immunopathogenesis

- In certain diseases, such as rheumatic fever and acute glomerulonephritis, it is not the organism itself that causes the symptoms of disease but the immune response to the presence of the organism.
 - ✓ In rheumatic fever, antibodies are formed against the M protein of *S. pyogenes*, which cross-react with joint, heart, and brain tissue.
 - ✓ Inflammation occurs, resulting in the arthritis, carditis, and chorea that are the characteristic findings in this disease
- Some bacterial infections are associated with cancers
- Cytokine Storm mechanisms

Typical Infectious Disease Stages

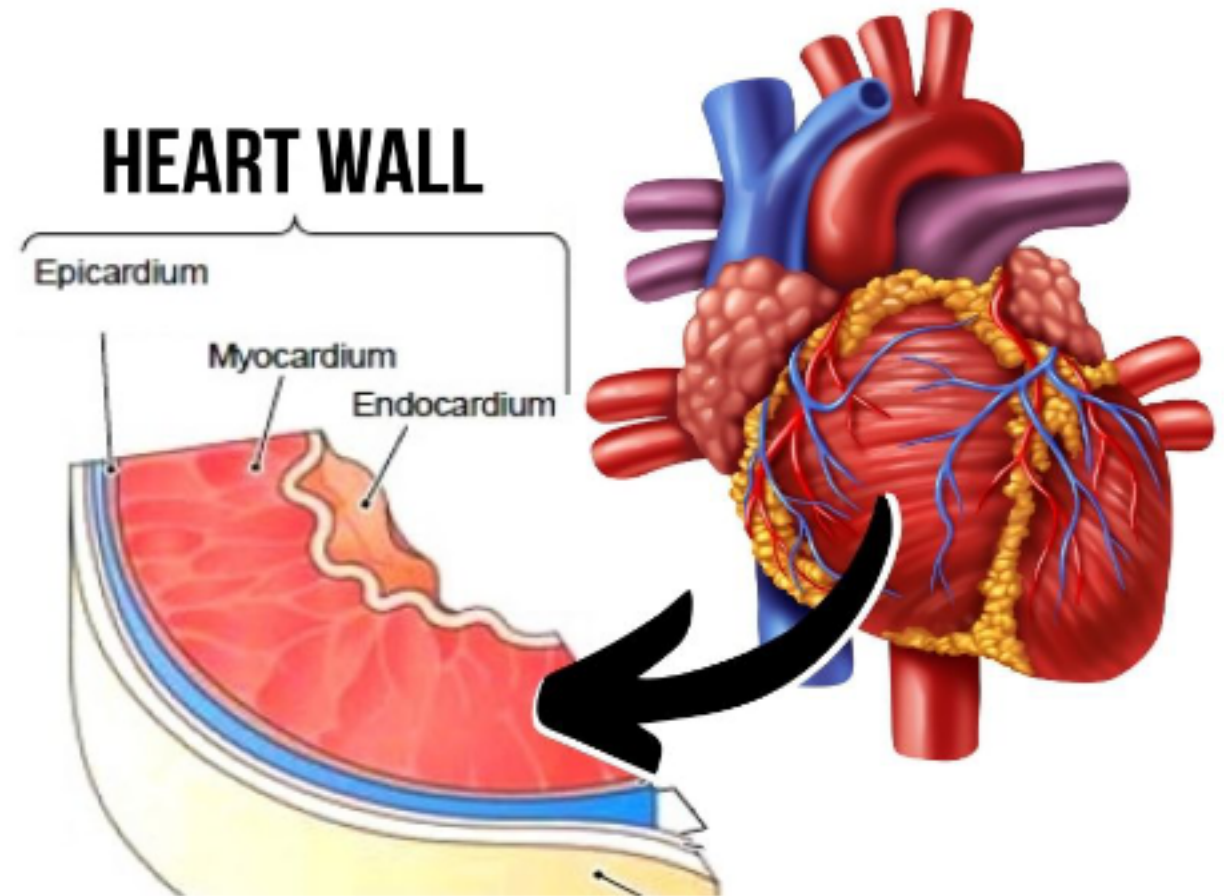
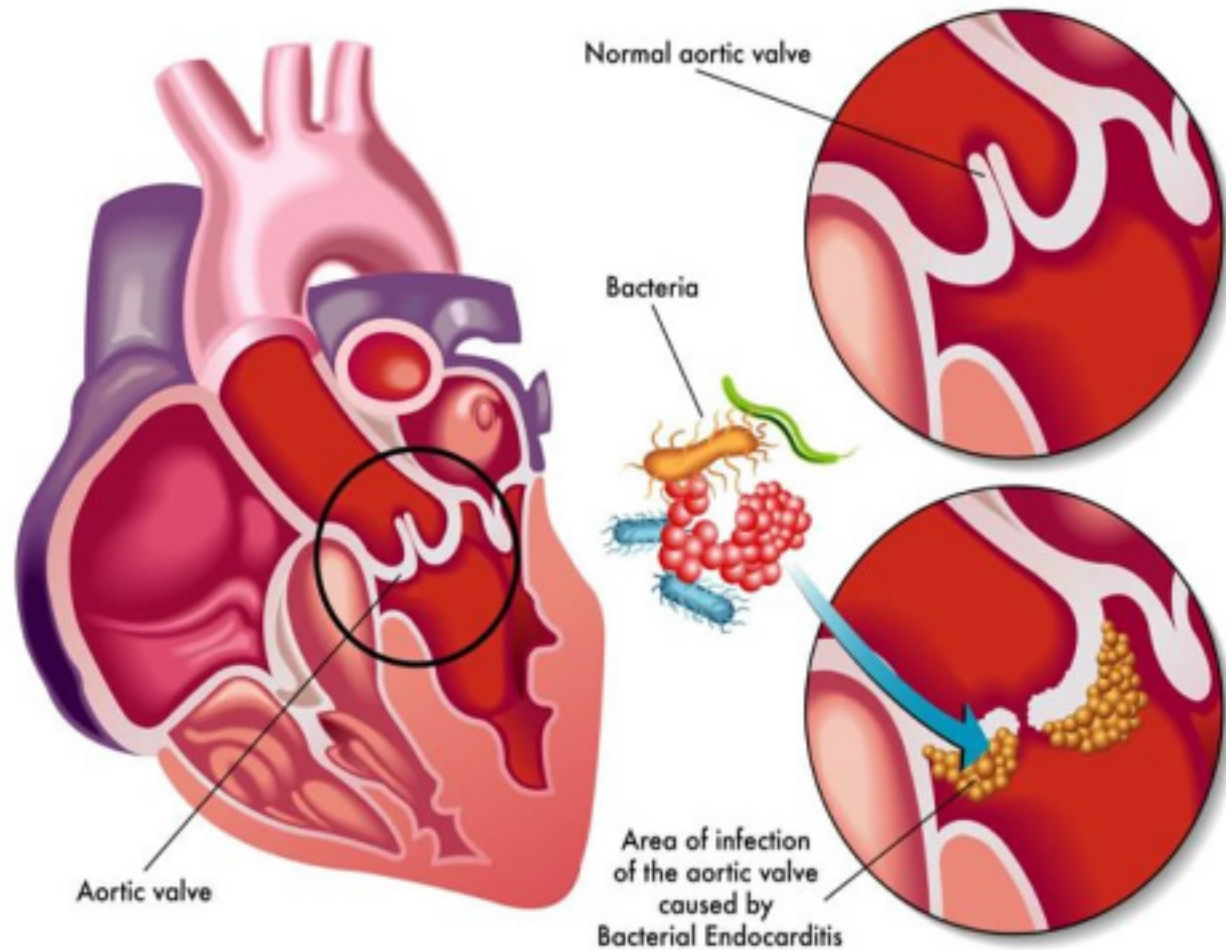
The Clinical Consequence: Stages of Infection



System-based Bacterial Pathogens

- Bacterial infections are best approached by site of infection (clinical syndrome) rather than memorizing organisms
- Where is the site, common organisms at this site, how is it transmitted and who is at risk
- Determinants of the likely pathogen, age and immune status of the host, community vrs hospital acquired
- We will now examine key bacterial pathogens based on each organ system and the diseases they cause

Cardiac bacterial disease – Endocarditis



Endocarditis – Pathogens

Category	Pathogen
Native valve	
Community onset	Viridans group streptococci, <i>Staphylococcus aureus</i> , <i>Streptococcus bovis</i> , <i>Enterococcus</i> species
Health care associated	<i>S. aureus</i> , <i>Enterococcus</i> species, <i>Staphylococcus epidermidis</i>
Intravenous drug user	<i>S. aureus</i> , gram-negative rods such as <i>Pseudomonas</i> , <i>Candida</i> species
Prosthetic valve	
Early	<i>S. epidermidis</i> , <i>S. aureus</i>
Late	<i>S. aureus</i> , viridans group streptococci, <i>Enterococcus</i> species, <i>S. epidermidis</i>
Pacemaker or defibrillator	<i>S. epidermidis</i> , <i>S. aureus</i>
Culture negative	Prior antibiotics, <i>Bartonella</i> species, <i>Coxiella burnetii</i> , <i>Brucella</i> species, <i>Tropheryma whipplei</i>

Endocarditis

- Diagnosis is made by
 - ✓ History and examination findings
 - ✓ Microbiological investigations – blood culture – at least 2 samples taken 12 hours apart
 - ✓ Radiological findings - Echocardiography
- Treatment is mainly medical usually with weeks of intravenous anti-biotics, occasionally, surgery may be required.

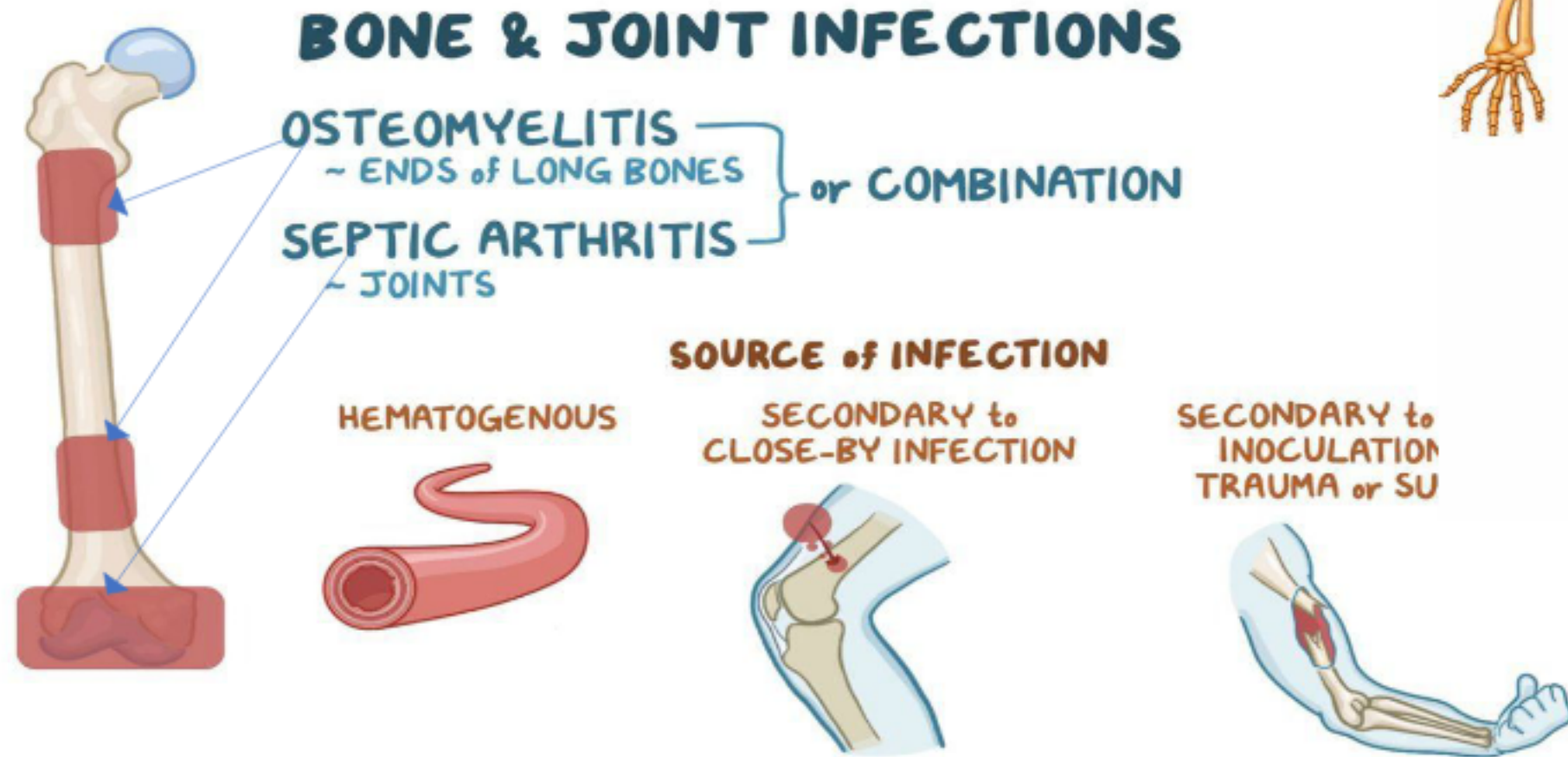
OTHER CARDIAC INFECTIONS

- Other cardiac infections include Pericarditis and Myocarditis
- Myocarditis
 - Viral pathogens are thought to be the predominant cause of infectious myocarditis, although many cases are idiopathic.
 - Coxsackie viruses are the most common cause, although cytomegalovirus, Epstein–Barr virus, parvovirus B19, and influenza virus have been implicated.
 - Other pathogens include *Trypanosoma cruzi*, the agent of Chagas' disease, and *Trichinella spiralis*
- Pericarditis refers to inflammation of the pericardium

Other Cardiac infections- Pericarditis

- *Mycobacterium tuberculosis* is one of the most common infectious causes of pericarditis worldwide.
- Other infectious causes of pericarditis include fungi and viruses
- Among viral infections, Coxsackie virus and echovirus are most common, although human immunodeficiency virus and cytomegalovirus can cause pericarditis as well.

BONE AND JOINT DISEASES



Bone and Joint Diseases – Osteomyelitis

- Clinical manifestation
 - ✓ The most characteristic clinical manifestations are bone pain and localized tenderness at the site of infection.
 - ✓ Most patients also have constitutional symptoms such as fever, night sweats, and fatigue
 - ✓ In acute osteomyelitis, the symptoms occur abruptly and progress rapidly, whereas in chronic osteomyelitis, the course is more indolent.
 - ✓ In chronic osteomyelitis, necrosis of the bone occurs.

Bone and Joint Diseases – Osteomyelitis

Organisms Causing Osteomyelitis with Various Predisposing Factors

Predisposing Factor	Common Organisms
Neonates	<i>Streptococcus agalactiae</i> (group B <i>Streptococcus</i>)
Children and adults	<i>Staphylococcus aureus</i>
Prosthetic joints	<i>Staphylococcus epidermidis</i>
Adults with vertebral osteomyelitis	<i>S. aureus</i> , <i>Mycobacterium tuberculosis</i>
Intravenous drug users	<i>S. aureus</i> , <i>Pseudomonas aeruginosa</i> , <i>Serratia marcescens</i> , <i>Candida albicans</i>
Skin infection in diabetic patient	<i>S. aureus</i> , anaerobes
Puncture wounds of foot	<i>P. aeruginosa</i>
Cat bite	<i>Pasteurella multocida</i>
Sickle cell anemia	<i>Salmonella</i> species

Bone and Joint Diseases – Osteomyelitis

- A microbiologic diagnosis of acute osteomyelitis is most consistently made by culture of a specimen (aspirate, biopsied tissue) of the bone lesion.
- Supportive blood cultures are positive in approximately half of cases.
- Radiologic imaging is mostly employed in diagnosing chronic osteomyelitis
- **Treatment**
 - ✓ involves use of bactericidal anti-biotics that penetrate soft tissue and bone for weeks to months eg Clindamycin
 - ✓ Surgical debridement is often necessary in chronic osteomyelitis

Bone and Joint Diseases – Septic Arthritis

- Infectious (septic) arthritis is an infection of the joints. The terms infectious or septic are used to distinguish these infections from immune-mediated arthritis, such as rheumatoid arthritis.
- Organisms typically reach the joint via the bloodstream from a skin site.
- Clinical manifestation
 - ✓ The acute onset of an inflamed joint. Fever is often present.
 - ✓ On physical examination, the affected joint is red, warm, and swollen, and a joint effusion (fluid in the joint) is typically present.
 - ✓ Reluctance to use a joint, especially in a child, may be a sign of infectious arthritis.

Bone and Joint Diseases – Septic Arthritis

Organisms Causing Infectious Arthritis with Various Predisposing Factors

Predisposing Factor	Common Organisms
Neonates	<i>Streptococcus agalactiae</i> (group B <i>Streptococcus</i>)
Children and adults	<i>Staphylococcus aureus</i>
Sexually active adults	<i>Neisseria gonorrhoeae</i>
Prosthetic hip and knee joints	<i>S. aureus</i> , <i>Staphylococcus epidermidis</i>
Intravenous drug users	<i>S. aureus</i> and <i>Pseudomonas aeruginosa</i>

Septic arthritis of knee



Bone and Joint Diseases – Septic Arthritis

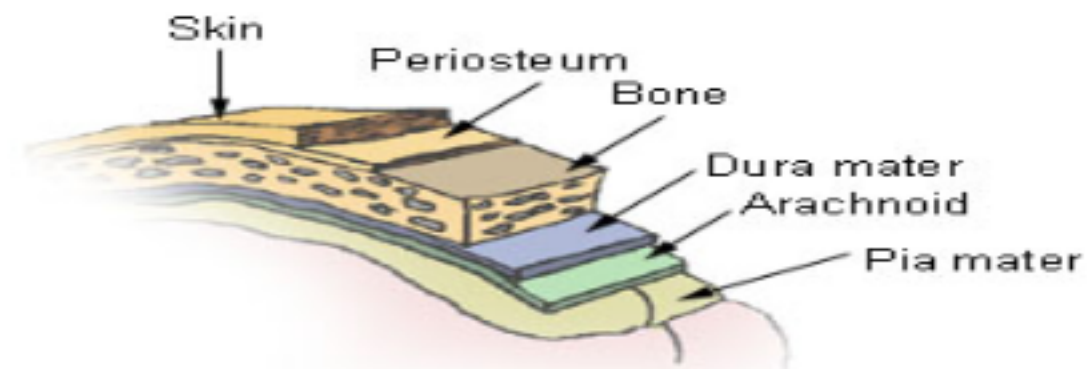
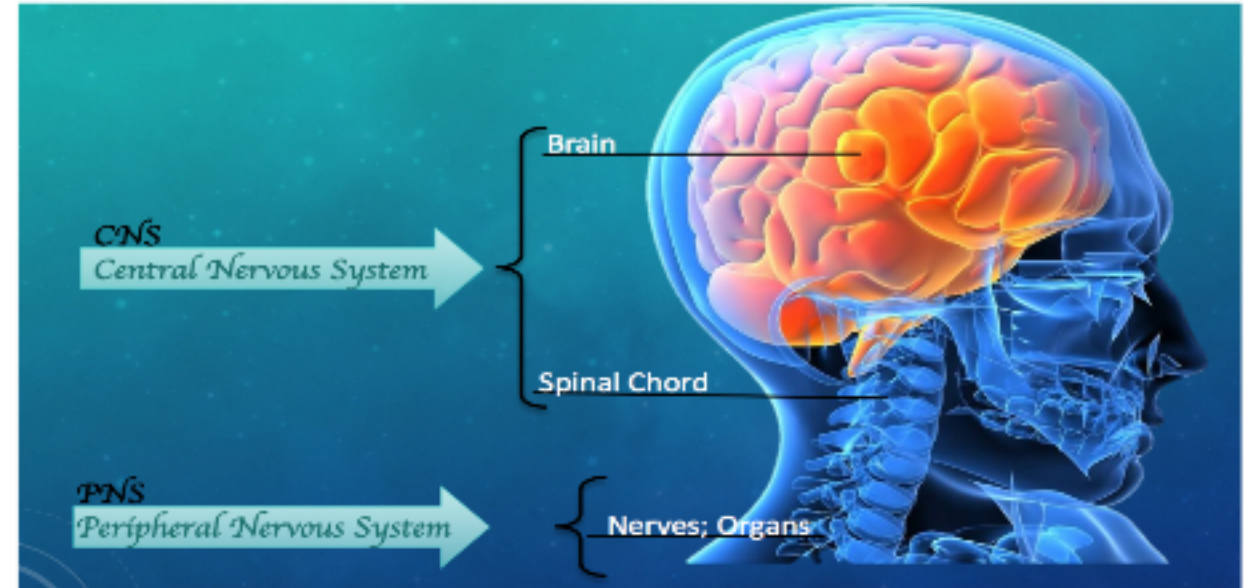
- Diagnosis
 - ✓ Visualization of the organisms in the Gram stain of joint fluid is used to guide empiric therapy.
 - ✓ A microbiologic diagnosis of infectious arthritis is typically made by culture of a specimen of the joint fluid. Blood cultures are positive in less than 30% of cases.
- Treatment
 - ✓ Prompt antibiotic treatment is required for optimal recovery
 - ✓ Untreated infectious arthritis can lead to joint destruction and loss of mobility
 - ✓ Removal of joint fluid via arthrocentesis and/or surgical drainage is an important adjunct to antibiotics.

Central Nervous System Infections

- Central nervous system (CNS) infections are often life threatening and can have severe sequelae.
- These infections cause inflammation and oedema within the unyielding cranium, resulting in damage to brain tissue and loss of function.
- Bacteria and viruses are the commonest etiologic agents, but fungi, protozoa, and helminths also cause these infections

Central Nervous System Bacterial Infections

- MENINGITIS
 - ✓ Inflammation of the meninges – the membranes that cover the brain and spinal cord.
 - ✓ Typically, bacterial meningitis reflects infection of the arachnoid mater and the CSF in both the Subarachnoid space and the ventricles



Meninges of the CNS

CNS INFECTION - MENINGITIS

- Acute Meningitis Pathogens
 - ✓ The most common bacterial cause of acute meningitis overall is *S. pneumoniae*.
 - ✓ However, *Streptococcus agalactiae* (group B Streptococcus) predominates in neonates, and *N. meningitidis* is common in teenagers and young adults
 - ✓ *Haemophilus influenzae* type B used to be an important cause in young children, but the widespread use of the conjugate polysaccharide vaccine has greatly decreased its incidence.
 - ✓ *Listeria monocytogenes* is reasonably common in the very young, the very old, and in immunocompromised patients and pregnant women
- Subacute or Chronic Meningitis
 - ✓ *Mycobacterium tuberculosis* is the leading cause of subacute/chronic bacterial meningitis

CNS INFECTION – BACTERIAL MENINGITIS

Predisposing Factor	Common Organisms
Neonate	<i>Streptococcus agalactiae</i> (group B <i>Streptococcus</i>), <i>Escherichia coli</i> , <i>Listeria monocytogenes</i>
Young children	<i>Streptococcus pneumoniae</i> , <i>Neisseria meningitidis</i> , <i>Haemophilus influenzae</i> ; also enteroviruses
Teenagers and young adults	<i>S. pneumoniae</i> , <i>N. meningitidis</i> ; also herpes simplex virus type 2
Older adults	<i>S. pneumoniae</i> , <i>N. meningitidis</i> , <i>L. monocytogenes</i>
Immunocompromised	<i>L. monocytogenes</i> , aerobic gram-negative rods (<i>Pseudomonas</i> and <i>Klebsiella</i>)
Pregnant women	<i>L. monocytogenes</i>
Cerebrospinal fluid leak; splenectomy; sickle cell anemia	<i>S. pneumoniae</i>
Deficiency of late-acting complement components; military recruits	<i>N. meningitidis</i>
After neurosurgery	<i>Staphylococcus aureus</i>
Ventriculoperitoneal shunt	<i>Staphylococcus epidermidis</i>

INVESTIGATORY TOOLS FOR CNS INFECTIONS

- History and examination are the first types available to the clinician in CNS infections investigation
- Radiological (mainly Computed Tomography Scan and Magnetic Resonance imaging), Microbiological and to some extent Serology are the mainstay of CNS infections investigations.
- The samples mostly utilised in microbiological or serological investigation of CNS infections are Cerebro-spinal Fluid (CSF) and blood.

CNS INFECTIONS - CSF ANALYSIS

- CSF obtained for analysis;
 - ✓ Gram stain
 - ✓ Bacterial culture
 - ✓ cells (for both cell type and number; neutrophils or lymphocytes),
 - ✓ Protein concentration,
 - ✓ Glucose concentration
 - ✓ Other appropriate tests (eg Rapid Serologic tests, PCR) depending upon concern for possible etiologies
- Analysis of spinal fluid can distinguish between acute bacterial meningitis and viral meningitis.

CSF ANALYSIS - MENINGITIS

GRAM STAIN

- ✓ Gram stain should be routinely performed on all CSF samples from patients with suspected Meningitis
- ✓ It has advantage of suggesting bacterial aetiology one day or more before culture results are available.
- Gram positive diplococci suggest pneumococcal infection
- Gram negative diplococci suggest meningococcal infection
- Small pleomorphic Gram negative coccobacilli suggest *Haemophilus influenzae* infection
- Gram positive rods and coccobacilli suggest Listerial infection
- ✓ Acid Fast Bacillus stain for TB meningitis

Gram Stain

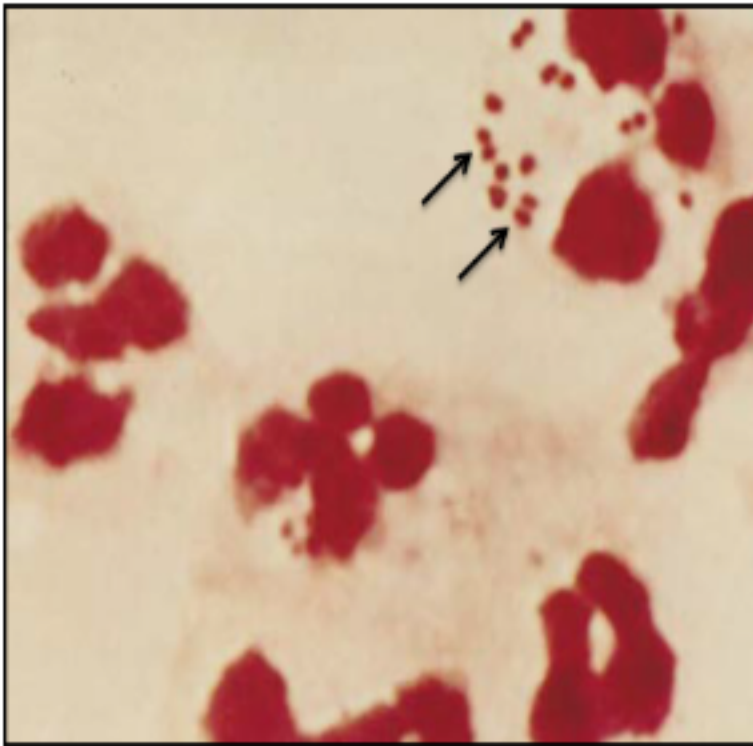


Figure 1. Gram stain of *N. meningitidis* in CSF with associated PMNs.

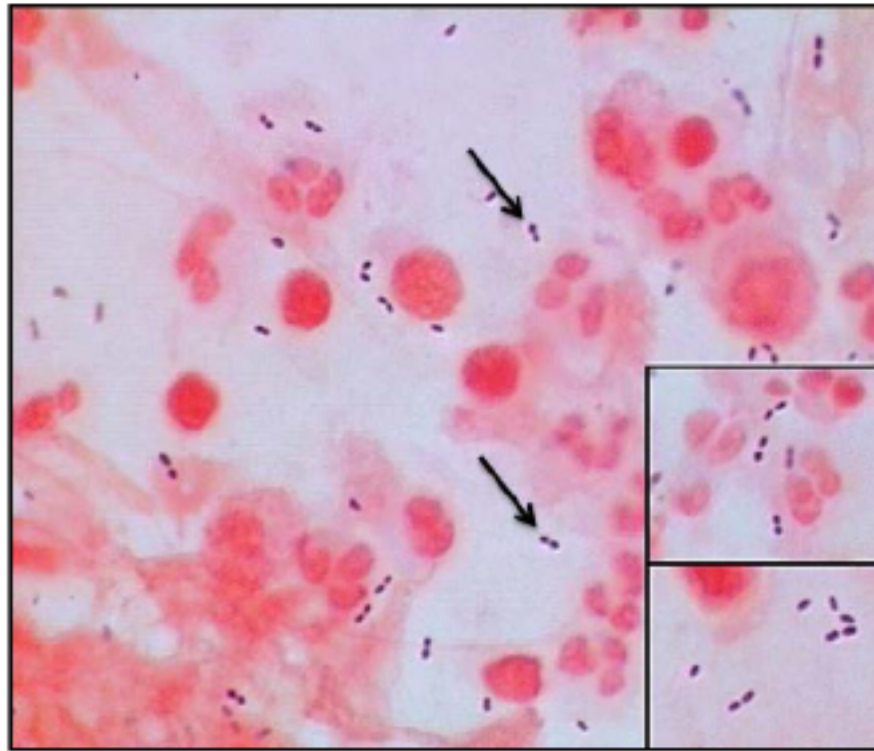


Figure 2. Gram stain of *S. pneumoniae* with WBCs

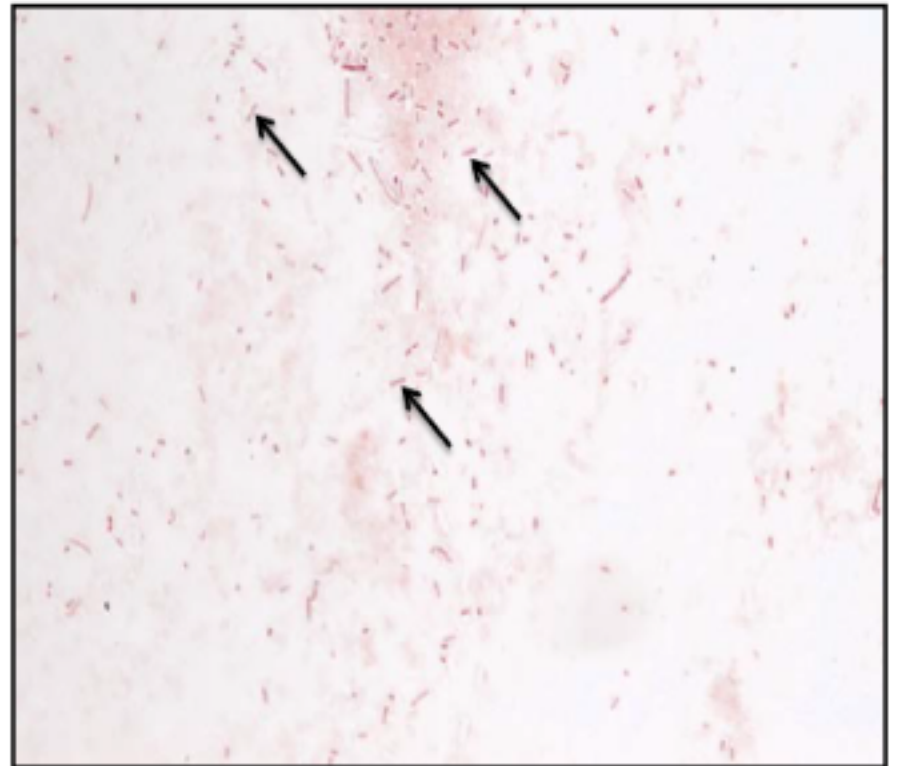


Figure 3. Gram stain of *H. influenzae*

PRESUMPTIVE DIAGNOSIS BACTERIAL MENINGITIS

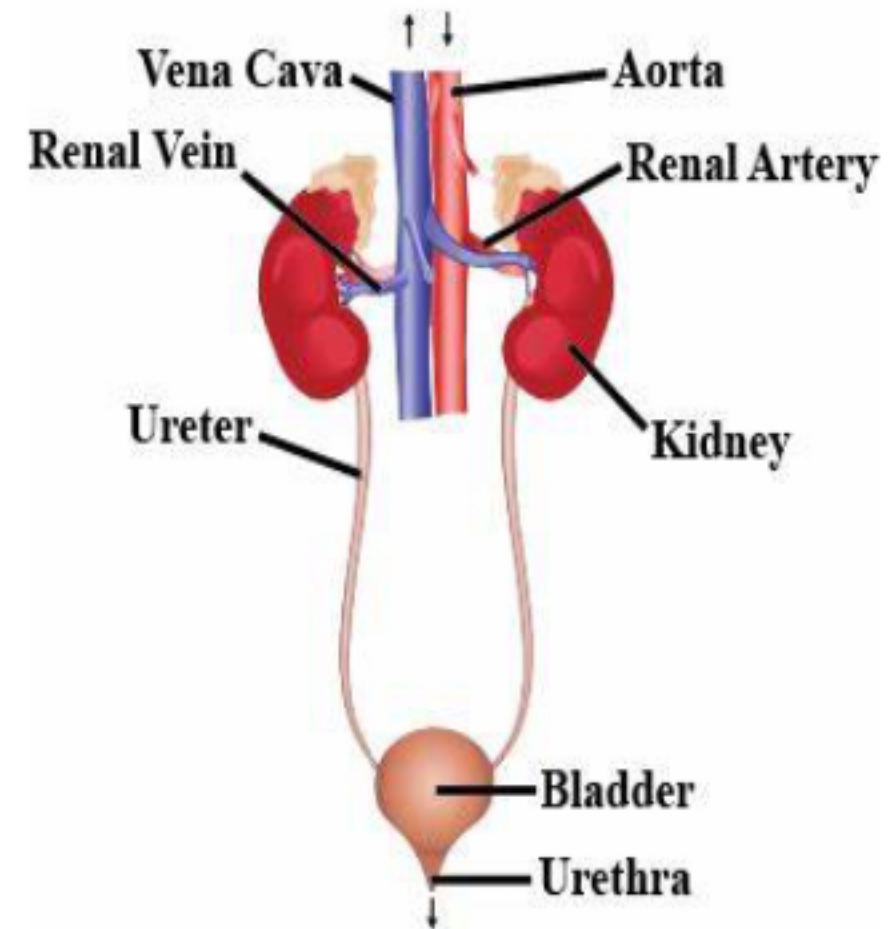
Presumptive Diagnosis

Table 1. Presumptive identification of *N. meningitidis*, *S. pneumoniae*, and *H. influenzae* based on growth on primary culture media and Gram stain results

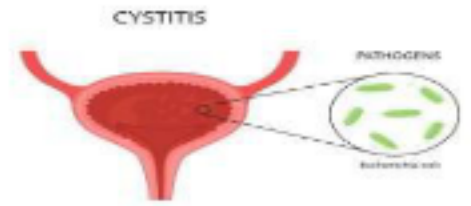
Growth on CAP	Growth on BAP	Gram stain	Presumptive ID
+	+	gram-negative diplococci	<i>N. meningitidis</i>
+	+	gram-positive diplococci	<i>S. pneumoniae</i>
+	-	gram-negative pleomorphic coccobacilli	<i>H. influenzae</i>

Urinary Tract Bacterial Infections

- Urinary tract infections are a group of common diseases that occur predominantly by ascension of normal enteric flora through the urethra into the bladder.
- These infections more frequently affect women due to anatomic differences including a shorter urethra.
- Cystitis is an infection of the bladder
- Pyelonephritis is an infection of the kidney(s). "Pyelo" refers to the renal pelvis, and "nephritis" means inflammation of the kidney.



Urinary Tract Infections



- Clinical manifestations of cystitis include dysuria (pain with urination); frequent, low volume urination; suprapubic tenderness and gross hematuria.
- Pyelonephritis typically present with fever, flank pain, nausea, and vomiting.
- *Escherichia coli* is by far the most common cause of urinary tract infections, especially cystitis.
- Other enteric gram-negative rods such as *Klebsiella* species and *Proteus* species are regular culprits.
- Gram-positive pathogens include *Enterococcus* species and *Staphylococcus saprophyticus*. *Staphylococcus saprophyticus* is common in young women.
- *Mycobacterium tuberculosis* may occasionally cause



Urinary Tract Infections

- **Prostatitis** is inflammation of the prostate, most often caused by bacterial infection.
- Infection most frequently occurs via the urethra then into to the prostatic ducts. However, hematogenous seeding of the prostate can occur as well
- Clinical Manifestations
 - ✓ Acute prostatitis may present with acute onset of fever, dysuria, urinary frequency, and severe pain with palpation of the prostate.
 - ✓ Patients may be very ill and can present with severe sepsis.
 - ✓ In contrast, chronic prostatitis presents with more subacute onset of dysuria, frequency, urinary hesitancy, and pelvic discomfort
- In younger patients, *Neisseria gonorrhoeae* and *Chlamydia trachomatis* are the most common causes of prostatitis
- Enteric bacteria, such as *E. coli*, are the predominant pathogens in older patients

Urinary Tract Infections - Diagnosis

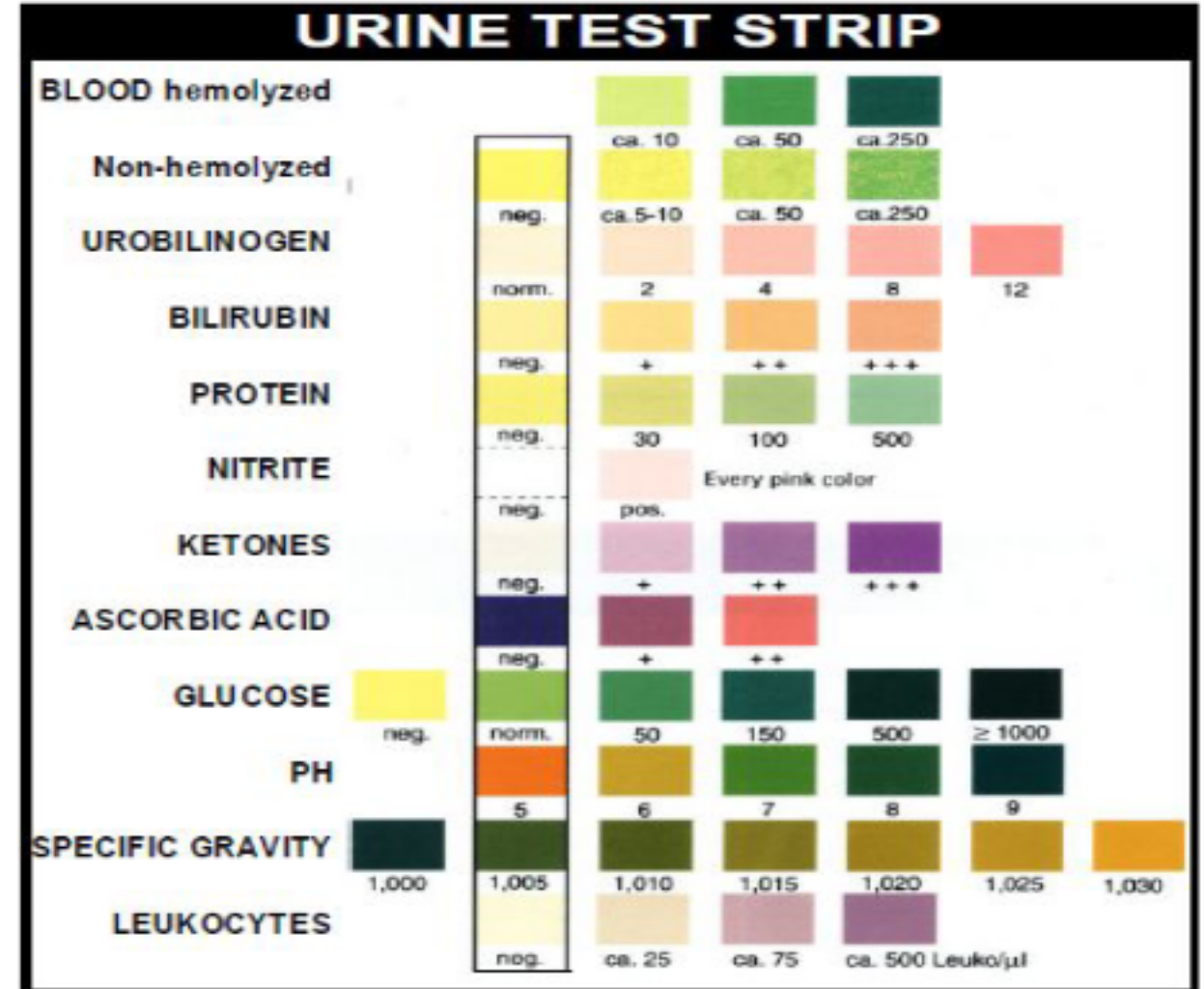
- **Urine microscopy**

- Microscope to look at urine.
- In patients with urinary tract infections, one can often find pyuria (elevated white blood cells [WBCs] in urine) and hematuria (red blood cells in urine), and sometimes bacteria can be seen.
- The presence of WBC casts indicates pyelonephritis rather than cystitis.
- A urine sample that has abundant squamous epithelial cells suggests that it is contaminated, and the results of the culture are not reliable.

- **Urine dipsticks**

- ✓ Different chemicals reagents on a strip that is dipped in urine to diagnose urinary tract diseases.
- ✓ Certain dipstick test results are suggestive of infection, namely positive leukocyte esterase and positive nitrite
- ✓ The positive nitrite occurs from the conversion of nitrate to nitrite by

URINARY TRACT INFECTIONS – Diagnosis dipstick



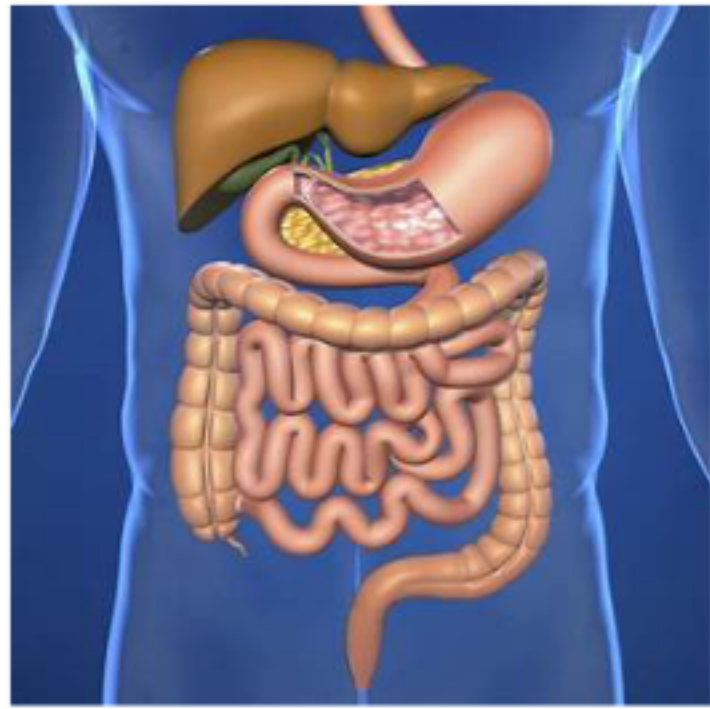
URINARY TRACT INFECTIONS- Diagnosis

- **Urine culture**

- ✓ allows identification of the organism causing infection.
- ✓ Urine in the bladder is normally sterile.
- ✓ Because contamination of samples can occur as urine passes through the outer third of the urethra, a numeric threshold of colony-forming units (CFUs) per milliliter has been established to confirm infection.
- ✓ In samples obtained from a midstream void, $\geq 1 \times 10^5$ CFU/mL is consistent with infection
- ✓ Either a voided midstream urine specimen or a specimen obtained by bladder catheterization can be used for urine culture

QUESTIONS

Gastrointestinal Tract Infections



- Infections with a variety of agents can occur in any part of the gastrointestinal (GI) tract from the mouth to the anal canal.
- Infections can range in severity from self-limited to life-threatening.
- Infections are typically caused by the ingestion of exogenous pathogens in sufficient quantities to evade host defences – disease is caused by multiplication, toxin production, or invasion through the gastrointestinal mucosa to reach the bloodstream and other tissues.

GASTROINTESTINAL TRACT INFECTIONS

Characteristics of Watery Diarrhoea compared with Bloody

D

Characteristics of Watery Diarrhea	Characteristics of Bloody Diarrhea
No red blood cells or white blood cells in stool, i.e., no inflammation	Typically both red blood cells and white blood cells in stool, i.e., inflammatory response
Typically afebrile	Often febrile
Usually large volume diarrhea	Usually small volume diarrhea
Infection typically in small intestine	Infection typically in colon

GASTROINTESTINAL TRACT INFECTIONS

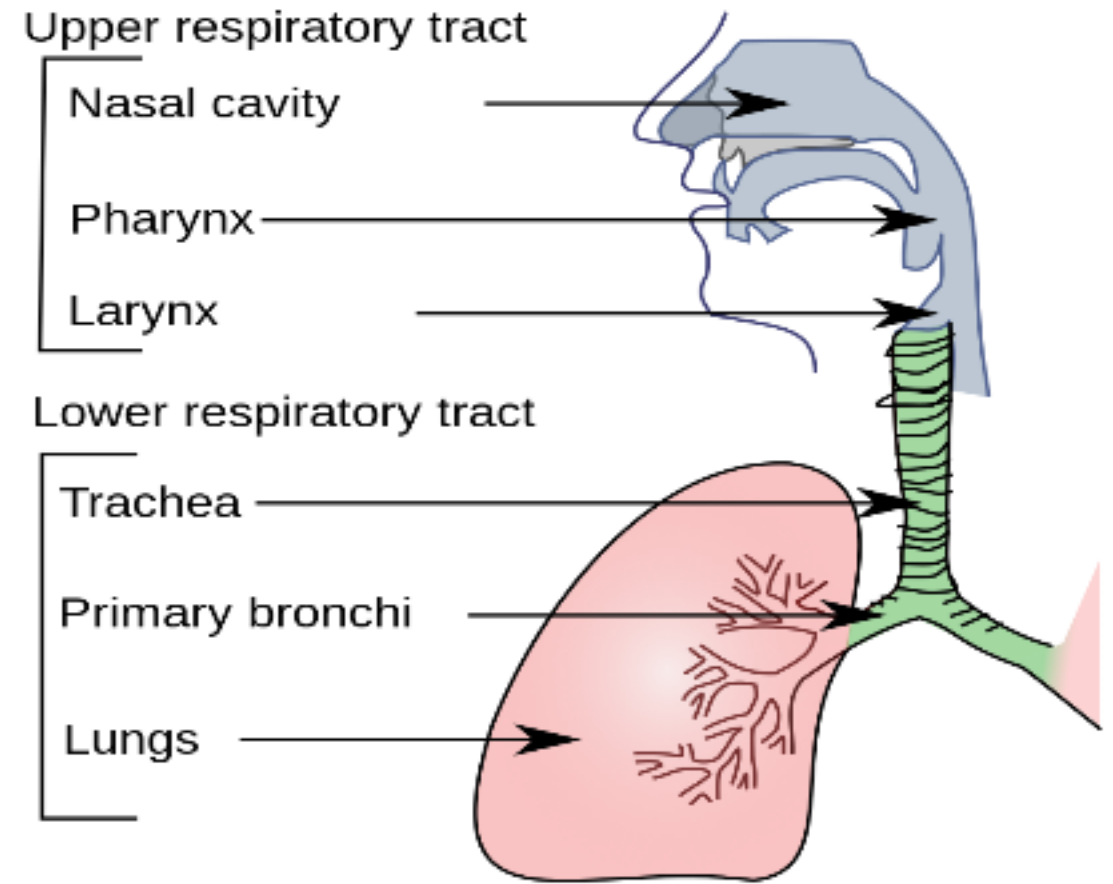
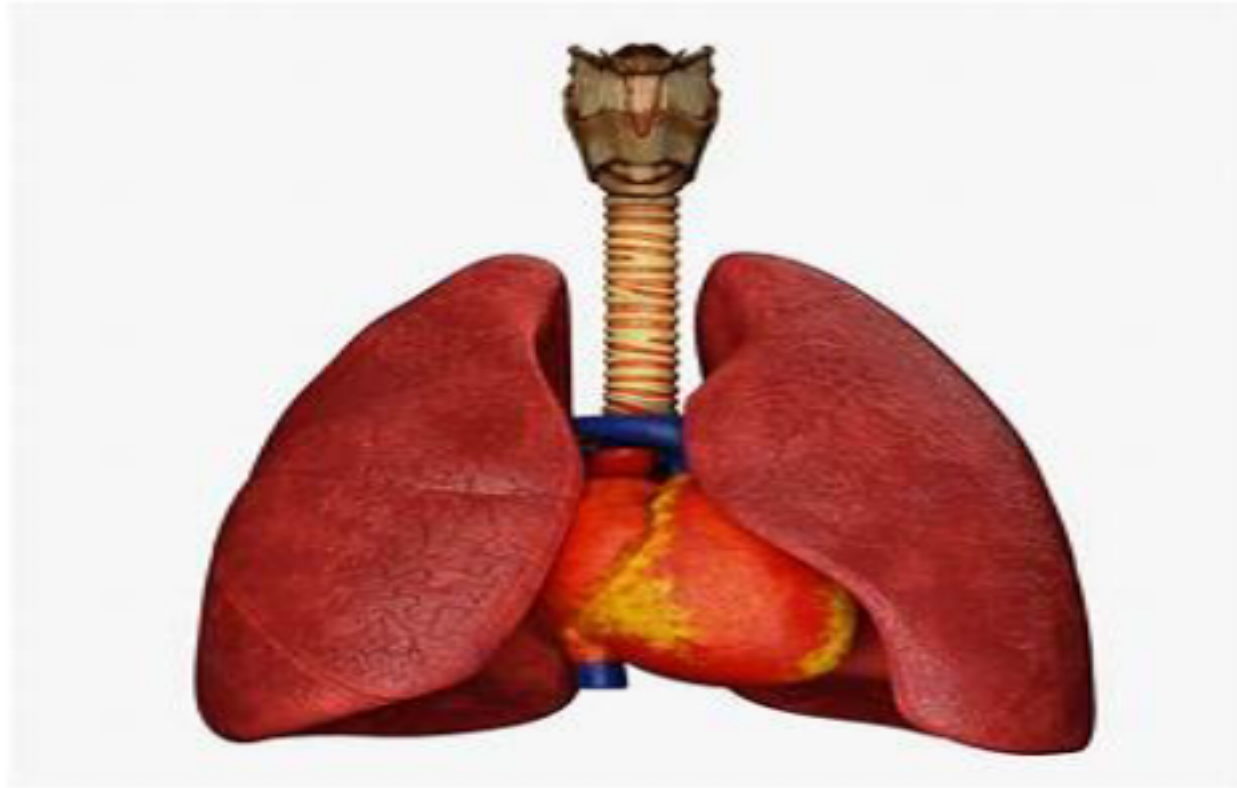
- Important Organisms That Typically Cause Either Watery or Bloody Diarrhoea

Organisms That Cause Watery Diarrhea	Organisms That Cause Bloody Diarrhea
Enterotoxigenic <i>Escherichia coli</i> (ETEC)	Shigatoxin-producing <i>Escherichia coli</i> (STEC)
<i>Vibrio cholerae</i>	<i>Shigella</i> species
<i>Staphylococcus aureus</i>	<i>Salmonella enterica</i>
<i>Bacillus cereus</i>	<i>Campylobacter jejuni</i>
<i>Listeria monocytogenes</i>	<i>Clostridium difficile</i>
Norovirus	<i>Yersinia enterocolitica</i>
Rotavirus	<i>Entamoeba histolytica</i>
<i>Giardia lamblia</i>	
<i>Cryptosporidium hominis</i>	

Gastrointestinal Bacterial Diseases

- Usually stool specimen is used in Gastrointestinal bacterial disease diagnoses
- *H. pylori* stool antigen test in Gastritis/PUD
- The culture medium is selected to suit the organism of interest
 - eg MacConkey for Enterobacteriaceae
 - Salmonella and Shigella Agar
 - Selenite F Broth for Salmonella enrichment
 - TCBS for *Vibrio Cholerae*
 - Skirrow medium for Campylobacter

Respiratory Infections



Upper Respiratory Tract Infection

- Bacterial aetiologies of some of the common upper respiratory tract infections may be primary or superinfections of the original viral processes
- Otitis media
 - ✓ Otitis media is an infection of the middle ear caused by either viruses or bacteria.
 - ✓ Any process that leads to eustachian tube obstruction can result in fluid retention and concomitant infection of the middle ear
 - ✓ Patients present with ear pain and pressure, often accompanied by an upper respiratory tract infection. In infants, the ear pain may manifest as ear pulling.
 - ✓ Common bacterial aetiologies include *Streptococcus pneumoniae* and *Haemophilus influenzae*

Upper Respiratory Tract Infection - Pharyngitis

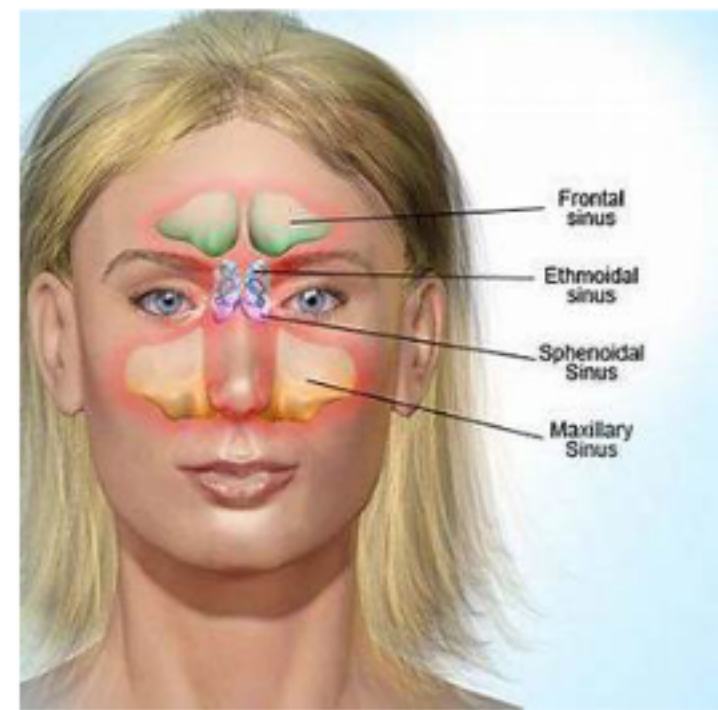
- ✓ About 10% of cases of pharyngitis are caused by *Streptococcus pyogenes* (group A *Streptococcus* [GAS]).
- ✓ Streptococcal pharyngitis (strep throat) is important because poststreptococcal immune sequelae, such as rheumatic fever, may occur
- ✓ Patients will complain of sore throat that is worse when swallowing. Fever may also be present
- ✓ *Streptococcus pyogenes* (GAS) is the most important bacterial cause
- ✓ Pharyngitis caused by *Neisseria gonorrhoeae* is likely to be the result of sexual activity and, if it occurs in children, is considered as a sign of child abuse.
- ✓ In certain countries where the diphtheria vaccine is not widely used, *Corynebacterium diphtheriae* is a significant cause of pharyngitis, often accompanied by a pseudo-membrane.

Upper Respiratory Tract Infection - Pharyngitis



Pharyngitis caused by Group A Beta Haemolytic Streptococcus

Upper Respiratory Tract Infection



- Sinusitis
 - ✓ Sinusitis is inflammation of the paranasal sinuses.
 - ✓ It can be either acute or chronic. Acute infections are considered those with symptoms lasting less than 4 weeks.
 - ✓ Impaired mucociliary clearance caused by viral infection or allergic rhinitis can obstruct the orifice of the sinus.
 - ✓ Clinical manifestations include purulent nasal discharge, nasal congestion, facial or sinus pain, decreased sense of smell, and fever. Headache and malodorous breath may be present.

UPPER RESPIRATORY TRACT INFECTION

- Epiglottitis
 - ✓ is an inflammation of the epiglottis.
 - ✓ Clinical manifestations include rapidly worsening sore throat and odynophagia (pain on swallowing) or dysphagia (difficulty in swallowing). Pain may be out of proportion to physical examination findings.
 - ✓ Airway obstruction can occur in severe cases. Epiglottitis in young children should be treated as a medical emergency.
 - ✓ *Haemophilus influenzae* type B is, by far, the most common cause, although the widespread use of the vaccine against *H. influenzae* type B has greatly reduced the incidence of epiglottitis.
 - ✓ Less common pathogens include other *H. influenzae* types, *S. pneumoniae*, *S. pyogenes*
 - ✓ Treatment is with anti-biotics; often intravenous Ceftriazone

UPPER RESPIRATORY TRACT INFECTION

- Other infections of the upper respiratory tract include
 - ✓ Common cold
 - Rhinoviruses (more than 100 serotypes) are the most common aetiology (up to 50%). Coronaviruses, adenoviruses, and enteroviruses such as Coxsackie viruses are other causes. Viruses such as parainfluenza virus and respiratory syncytial virus are also possible causes of the common cold, although they primarily cause other diseases (croup and bronchiolitis, respectively)
 - ✓ Croup
 - inflammation of the larynx, trachea, and large bronchi (laryngotracheobronchitis)
 - Parainfluenza viruses, especially type 1, are the most common cause. Respiratory syncytial virus and influenza virus account for 1% to 10% of cases.
 - ✓ Laryngitis
 - Laryngitis is inflammation of the vocal folds of the larynx
 - Parainfluenza viruses and rhinoviruses are the most common causes of laryngitis
 - Bacteria such as *S. pyogenes*, *M. catarrhalis*, and *H. influenzae* have also been isolated.

UPPER RESPIRATORY TRACT INFECTION

Common Infections of the Upper Respiratory Tract

Infection	Important Pathogens	Treatment
Otitis media	<i>Streptococcus pneumoniae</i> , <i>Haemophilus influenzae</i> , <i>Moraxella catarrhalis</i>	Amoxicillin
Acute sinusitis	<i>S. pneumoniae</i> , <i>H. influenzae</i> , <i>M. catarrhalis</i>	Amoxicillin if symptoms persist for >10 days
Pharyngitis	<i>Streptococcus pyogenes</i> (group A <i>Streptococcus</i>), viruses (e.g., adenovirus)	Penicillin or amoxicillin if group A <i>Streptococcus</i> diagnosed
Common cold	Rhinovirus, coronavirus, and others	Supportive; zinc may be helpful in reducing duration of symptoms
Croup	Parainfluenza virus	Supportive; corticosteroids and epinephrine if moderate or severe symptoms
Laryngitis	Parainfluenza virus and rhinovirus	Supportive
Epiglottitis	<i>H. influenzae</i> type B	Ceftriaxone

Infections Of The Lower Respiratory Tract

- Bronchitis

- ✓ Bronchitis is a self-limited inflammation of the bronchi.
- ✓ Acute bronchitis must be distinguished from chronic bronchitis in which patients have a cough for more than 3 months.
- ✓ Respiratory viruses are the most common pathogens
- ✓ Bacterial pathogens are not thought to play a significant role in acute bronchitis

- Bronchiolitis

- ✓ Bronchiolitis is inflammation of the bronchioles—the small airways less than 2 mm in diameter.
- ✓ RSV is the most common pathogen.
- ✓ Other aetiologies include influenza virus, parainfluenza virus, adenovirus, coronavirus, rhinovirus, and human metapneumovirus.
- ✓ Viruses are the main aetiology of bronchiolitis. Bacteria are not thought to

INFECTIONS OF THE LOWER RESPIRATORY TRACT

- Pneumonia – Pathogens

Bacterial and Viral Causes of Community-Acquired Pneumonia by Age (Listed in Order of Frequency)

Age	Bacteria	Viruses
Neonates	Group B streptococci <i>Escherichia coli</i>	Respiratory syncytial virus (RSV)
Infants	<i>Chlamydia trachomatis</i> <i>Streptococcus pneumoniae</i>	RSV Parainfluenza virus
Children	<i>S. pneumoniae</i> <i>Haemophilus influenzae</i>	RSV Parainfluenza virus
Young adults	<i>Mycoplasma pneumoniae</i> <i>Chlamydophila pneumoniae</i> <i>S. pneumoniae</i>	Various respiratory viruses (e.g., adenovirus)
Older adults	<i>S. pneumoniae</i> <i>H. influenzae</i> <i>Legionella pneumophila</i>	Influenza virus

INFECTIONS OF THE LOWER RESPIRATORY TRACT

Predisposing Factors Associated With Typical Pathogens Causing Community-Acquired Pneumonia

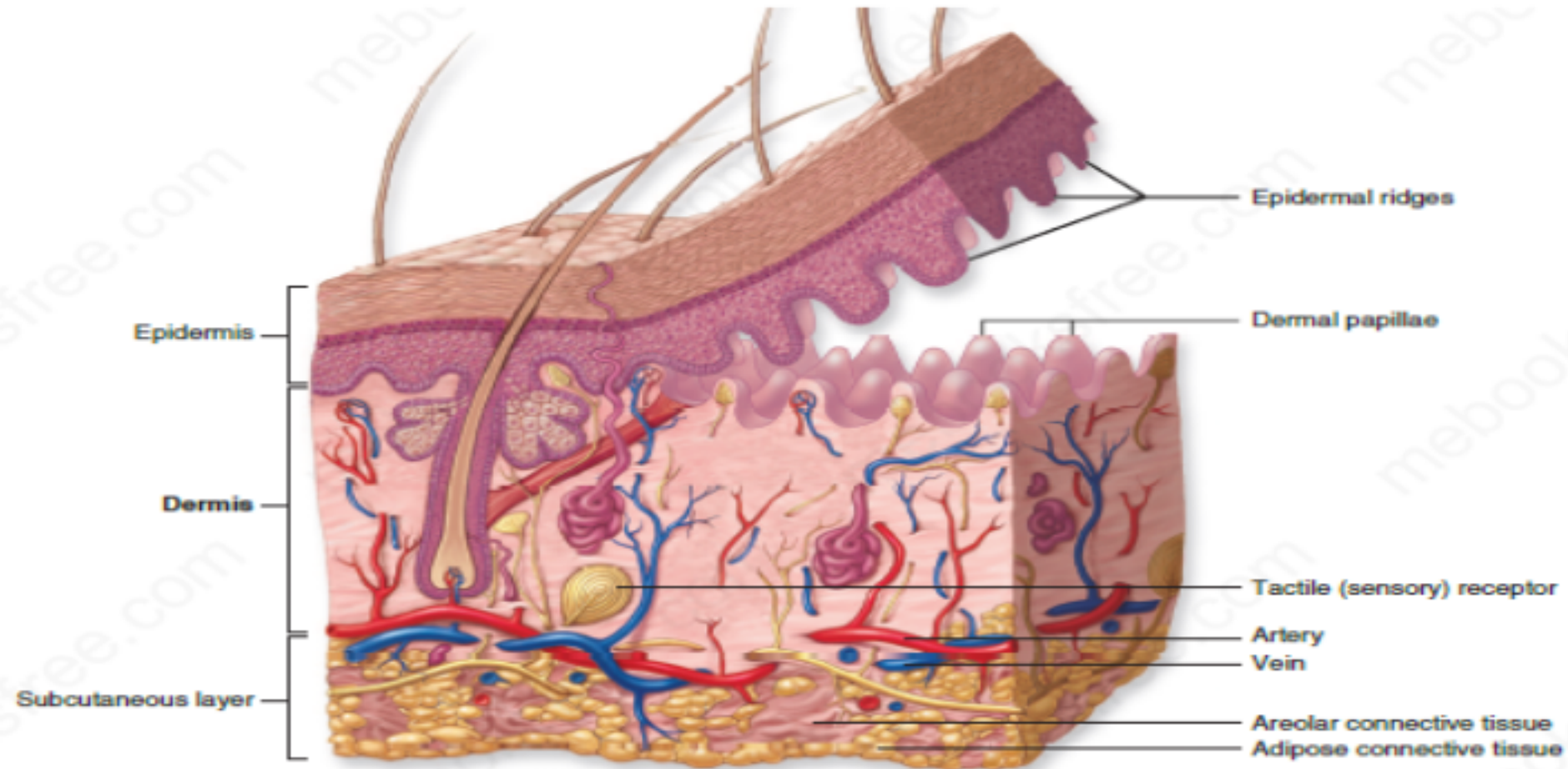
Predisposing Factors	Typical Pathogens
Alcoholism	<i>Klebsiella pneumoniae</i> , oral anaerobes
Bird exposure, especially psittacine birds such as parrots (psittacosis)	<i>Chlamydophila psittaci</i>
Chronic obstructive pulmonary disease (COPD), including smoking-related	<i>Haemophilus influenzae</i>
Cystic fibrosis	<i>Pseudomonas aeruginosa</i>
Imported wool, spores in wool (wool-sorter's diseases)	<i>Bacillus anthracis</i>
Influenza virus infection	<i>Staphylococcus aureus</i>
Intubation, postsurgery, and intensive care unit (ICU)	Coliforms, ¹ <i>P. aeruginosa</i> , <i>S. aureus</i>
Mouse droppings exposure, especially in southwestern states	Hantavirus
Sheep exposure, especially placental tissue (Q fever)	<i>Coxiella burnetii</i>
Travel to or reside in Central Valley of California, Arizona, or New Mexico	<i>Coccidioides immitis</i>
Travel to or reside in Ohio or Mississippi river valleys	<i>Histoplasma capsulatum</i>
Ventilator-associated, especially in ICU	<i>Acinetobacter</i> species
Water aerosols, especially from air conditioners	<i>Legionella pneumophila</i>

¹Coliforms such as *Escherichia coli*, *Klebsiella*, *Enterobacter*, *Serratia*, and *Proteus*.

INFECTIONS OF THE LOWER RESPIRATORY TRACT

- Pneumonia – Diagnosis
 - ✓ The “gold standard” for a diagnosis of pneumonia is an infiltrate on a plain chest radiograph
 - ✓ Sputum analysis for Gram stain and culture and blood cultures may be helpful in the hospitalized patient
 - ✓ Pneumonia caused by *M. tuberculosis* is diagnosed by acid-fast stain of sputum and culture on mycobacterial medium. A PCR assay done directly on sputum is available also.
 - ✓ Pneumonia caused by *Legionella pneumophila* is often diagnosed by urinary antigen.
 - ✓ PCR tests for various respiratory pathogens such as *M. pneumoniae*, *C. pneumoniae*, influenza virus, and RSV are useful in special circumstances.
 - ✓ Treatment is mainly anti-biotics tailored towards the possible

Skin And Soft Tissue Infections

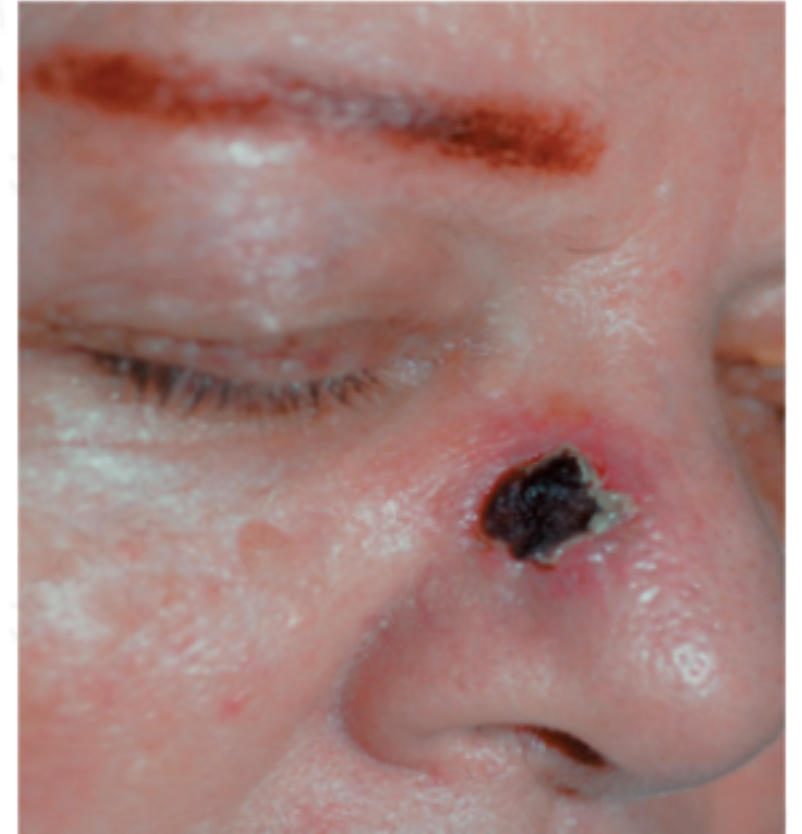


Layers of the skin and subcutaneous tissue

Skin And Soft Tissue Infections

- Impetigo
 - ✓ Impetigo is an infection of the epidermal layer of skin
 - ✓ There are two modes of acquisition of impetigo, primary infection that occurs in otherwise normal skin or secondary impetigo, which occurs following a break in normal skin integrity. Bacteria invade into the epidermal layer and cause local damage.
 - ✓ *Staphylococcus aureus* and *S. pyogenes* are the two main pathogens that cause impetigo.
 - ✓ In neutropenic patients, a clinical syndrome termed ecthyma gangrenosum which may be confused with impetigo is due to disseminated *Pseudomonas aeruginosa* infection.

Skin And Soft Tissue Infections



Skin And Soft Tissue Infections

- Cellulitis and Erysipelas
 - ✓ Cellulitis and erysipelas are infections of the dermis.
 - ✓ These infections occur following a break in normal skin integrity.
 - ✓ Cellulitis and erysipelas both involve the dermis, but erysipelas involves the upper dermis and superficial lymphatics and sharply demarcated, whereas cellulitis involves the deeper dermis and subcutaneous fat and lesion is not sharply demarcated
 - ✓ Beta-hemolytic streptococci are the most common pathogens to cause these infections, with *S. pyogenes* and *Streptococcus agalactiae* being some of the most common species.
 - ✓ *Staphylococcus aureus* can also cause this type of infection.

Skin And Soft Tissue Infections

Erysipelas. Note markedly inflamed lesion with clearly demarcated border on right cheek, across the nose to the left cheek.



Cellulitis. Note diffuse inflammation on right leg including the skin over the ankle and dorsum of the foot.



Skin And Soft Tissue Infections

- Folliculitis
 - ✓ Folliculitis is a superficial infection of the hair follicles
 - ✓ Bacteria and purulent material accumulate in hair follicles in the epidermal layer of the skin.
 - ✓ Folliculitis presents with pinpoint erythema around individual hair follicles. A small amount of purulence may be seen.
 - ✓ This can be seen in an isolated body area or throughout the skin
 - ✓ *Staphylococcus aureus* is the most common cause of folliculitis.
 - ✓ *Pseudomonas aeruginosa* can also cause folliculitis and is associated with the use of unchlorinated hot tubs.

Folliculitis

Diagnosis is made clinically, but if purulent material is present, it can be cultured.

Treatment

Folliculitis often resolves on its own, and treatment is not needed. Warm compresses or topical antibiotics can be considered in selected cases

Prevention

Handwashing and covering draining lesions should be used to prevent spread of bacteria.

Avoiding unchlorinated hot tubs is recommended



SKIN AND SOFT TISSUE INFECTIONS

- Skin Abscess (Furuncle and Carbuncle)
 - ✓ A skin abscess is an infection of the dermis and deeper layers of skin that contains purulent material
 - ✓ Abscesses occur when pathogens enter a break in the skin following trauma or when they spread from infected hair follicles.
 - ✓ When a single follicle is infected and tracks down into the dermis, it is termed a furuncle ("boil"), and when multiple infected hair follicles coalesce, it is termed a carbuncle.
 - ✓ Occasionally an abscess may develop following hematogenous dissemination of an infection.

SKIN AND SOFT TISSUE INFECTIONS

- Skin Abscess (furuncle and carbuncle)
 - ✓ A carbuncle is a larger, more serious lesion than a furuncle. It is composed of several adjacent furuncles that have coalesced into an inflamed, indurated lesion that typically extends deep into subcutaneous tissue.
 - ✓ *Staphylococcus aureus* is, by far, the most common cause of skin abscesses (more than 75% of cases).
 - ✓ Gram stain and culture of purulent material obtained from the abscess allows for diagnosis
 - ✓ The primary treatment for abscesses is incision and drainage

SKIN AND SOFT TISSUE INFECTIONS

Abscess. Note localized area of inflammation containing a central core of yellowish pus (arrow) on medial aspect of foot.



Carbuncle. Note multiple furuncles that have coalesced to form a large area of inflammatory lesion



SKIN AND SOF TISSUE INFECTIONS

- Necrotizing Soft Tissue Infections (Necrotizing Fasciitis/Myonecrosis)
 - ✓ Necrotizing fasciitis is a necrotizing infection of the deep structures of the skin including the underlying fascia.
 - ✓ In myonecrosis, the underlying muscle becomes necrotic
 - ✓ Maybe polymicrobial (more common) or monomicrobial.
 - ✓ Monomicrobial infection is most often due to *S. pyogenes* but can also be caused by *Vibrio vulnificus* following trauma in brackish water
 - ✓ Necrotizing soft tissue infections are a medical emergency, and treatment requires a combination of surgical debridement of infected tissue and antibiotic therapy.

SKIN AND SOFT TISSUE INFECTIONS

Necrotizing fasciitis. Note two large hemorrhagic

bullae surrounded by dusky red inflamed tissue



Fournier's gangrene. Note gangrene of the

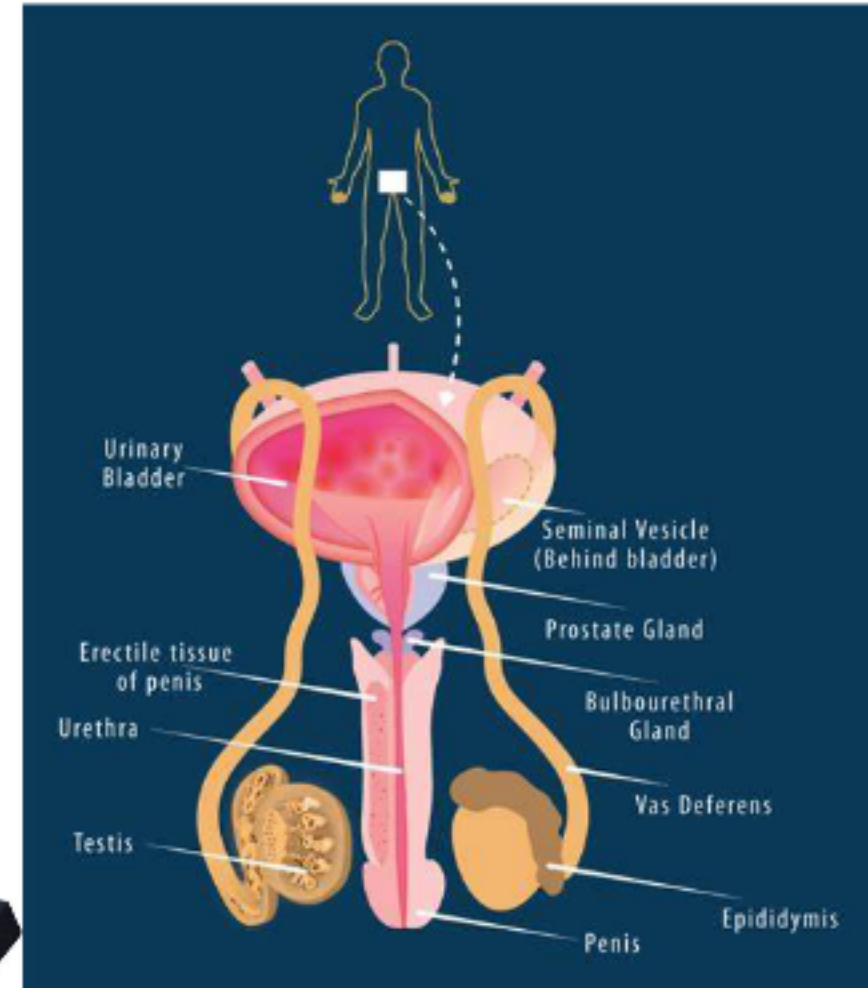
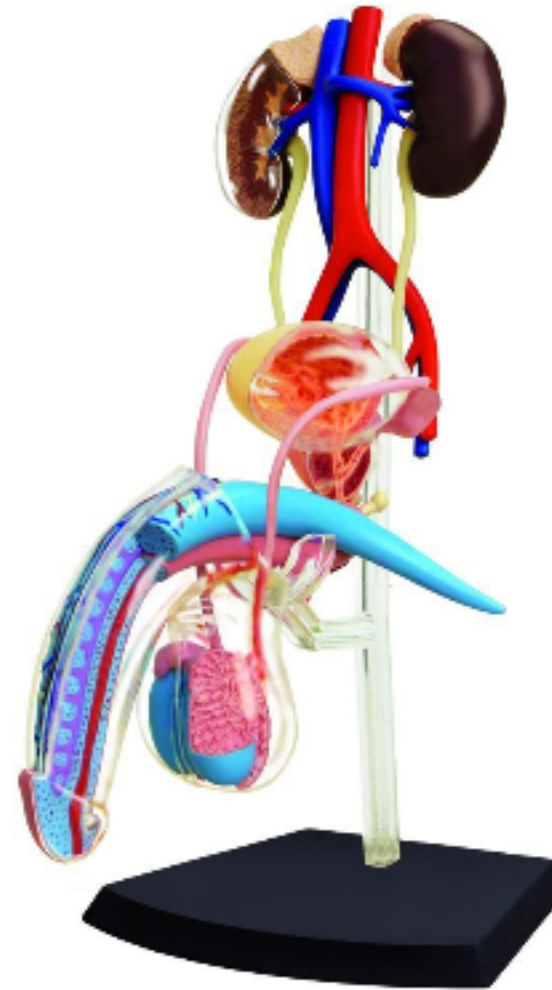
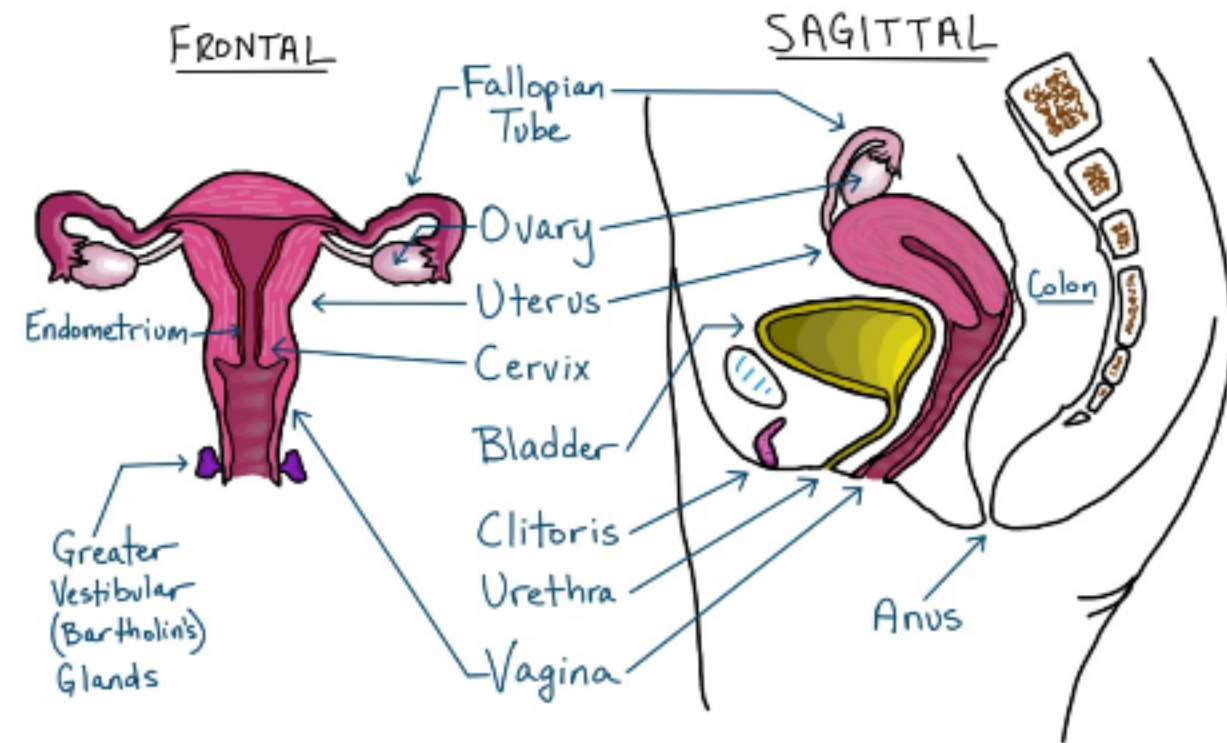
genitalia and skin of lower abdomen.



PELVIC INFECTIONS

- Infections in the pelvic organs and surrounding structures comprise a heterogenous group of diseases.
- They primarily affect sexually active women and men.
- Many of the pathogens implicated are sexually transmitted, so an important aspect of management is partner notification and treatment, as well as patient education regarding safe sexual practices.

PELVIC INFECTIONS



Reproductive Tract Diseases-Genital tract ulcers

- ✓ Genital ulcer disease manifests as a breach in the skin or mucosa of the genitalia, usually caused by a sexually transmitted infection.
- ✓ Herpes simplex virus type 2 (HSV-2) is the most common aetiology in most geographic areas, followed by syphilis and chancroid.
- ✓ HSV-2 (causes genital herpes), *Treponema pallidum* (causes primary syphilis/chancere), and *H. ducreyi* (causing chancroid).
- ✓ Less common pathogens include *Chlamydia trachomatis* (causing lymphogranuloma venereum) and *Klebsiella granulomatis* (causing granuloma inguinale, also known as

Reproductive Tract Diseases -Genital Ulcers



Genital herpes (HSV2)



Chancre(Primary Syphilis)

Syndrome	Pathogen	Appearance of Ulcer	Pain	Adenopathy	Diagnosis	Treatment
Genital herpes	HSV-2 (more common than HSV-1)	Multiple, small vesicles and ulcers with an erythematous base	Painful	Tender lymphadenopathy	Direct fluorescent antibody (DFA) testing and/or viral culture; sample taken from the base of the ulcer; Tzanck smear shows multinucleated giant cells; NAAT may be useful	Acyclovir, famciclovir, valacyclovir
Syphilis	<i>Treponema pallidum</i>	Single (usually), indurated ulcer with a clean base; self-resolves and may not be observed	Painless	Painless, regional lymphadenopathy; Lymph nodes feel "rubbery"	Serologic screening with non-treponemal test (e.g., rapid plasma reagin [RPR] or Venereal Disease Research Laboratory [VDRL]); treponemal test (e.g., fluorescent treponemal antibody absorption [FTA-ABS]) for confirmation; dark field examination of fluid from the lesion if possible	Benzathine penicillin G
Chancroid	<i>Haemophilus ducreyi</i>	Multiple, nonindurated ulcers with a gray or yellow exudate at the base	Very painful	Tender, unilateral inguinal lymphadenopathy; lymph nodes may rupture	Difficult to diagnose; special culture media, if available, or polymerase chain reaction (PCR)	Azithromycin
Lymphogranuloma venereum	<i>Chlamydia trachomatis</i> serovars L1–3	Small, shallow ulcers that self-resolve and are not usually observed	Painless	Characteristic appearance of lymph nodes is key feature; may be bilateral, large, and painful; presents with fluctuant "buboes" and sinus tracts	Nucleic acid amplification test (NAAT); serology can also be used	Doxycycline
Granuloma inguinale (donovanosis)	<i>Klebsiella granulomatis</i>	Marked, beefy red, vascular ulcer with granulomatous appearance and rolled edges	Painless	Not a major feature; subcutaneous granulomas, "pseudobuboes" may occur	Difficult to diagnose; PCR if available; may also see dark-staining oval organisms on Giemsa staining (Donovan bodies) of biopsy specimen	Doxycycline

Reproductive Tract Diseases

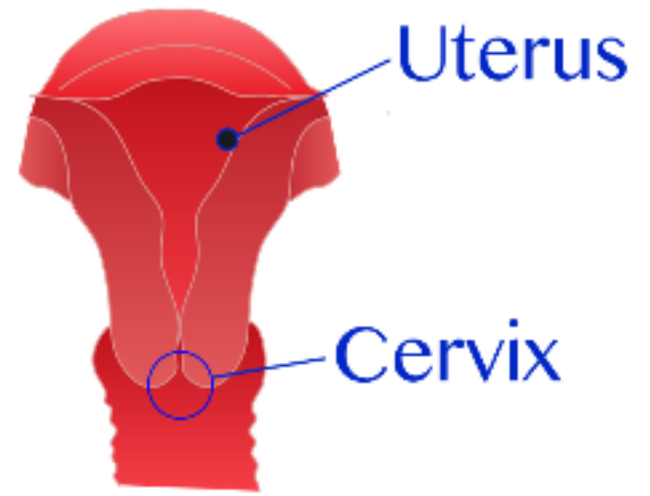
- Vaginitis
 - ✓ Vaginitis is inflammation of the vagina that can result in discharge, itching, and pain.
 - ✓ Common causes of vaginitis are candidiasis, trichomoniasis, and bacterial vaginosis.
 - ✓ Bacterial vaginosis is a polymicrobial infection, it does not appear to be a sexually transmitted disease.
 - ✓ Trichomoniasis, on the other hand, is a sexually transmitted disease.

Reproductive Tract Diseases - Vaginitis

Vaginitis: Clinical Features, Diagnosis, and Treatment

Syndrome	Pathogen(s)	Nature of Vaginal Discharge	pH	Microscopy	Treatment	Other Notes
Normal	<i>Lactobacillus</i> species are the predominant normal flora	Could be clear and mucoid, especially at midcycle estrogen surge; during pregnancy, secretions may be thicker and white	<4.5			
Bacterial vaginosis	<i>Gardnerella vaginalis</i> ; anaerobes such as <i>Mobiluncus</i> species are also involved	Malodorous, gray, thin	>4.5	Clue cells are epithelial cells covered with gram-variable bacteria (see Figure 74-6)	Metronidazole (orally or gel) or tinidazole (orally)	An amine-like "fishy" odor occurs after the addition of 10% potassium hydroxide
Vaginal candidiasis	<i>Candida albicans</i>	"Cottage cheese"; white and clumpy	<4.5	Yeasts and pseudohyphae seen in KOH prep (10% potassium hydroxide) (see Figure 74-7)	Fluconazole (orally), miconazole (vaginal suppository), butoconazole (vaginal cream) are single-dose options	
Trichomoniasis	<i>Trichomonas vaginalis</i>	Malodorous, green-yellow, thin	>4.5	Numerous neutrophils	Metronidazole or tinidazole (orally)	"Strawberry cervix" seen on speculum examination

Reproductive Tract Diseases



- Cervicitis

- ✓ Cervicitis is inflammation of the uterine cervix.
- ✓ Acute cervicitis is usually due to a sexually transmitted infection caused by either *C. trachomatis* or *Neisseria gonorrhoeae* or both.
- ✓ A large proportion of women with cervicitis are asymptomatic, in many cases, cervicitis is detected on speculum examination following routine screening
- ✓ On physical examination, increased friability of the cervical tissue after a swab is inserted may be a clue to the diagnosis.
- ✓ The usual pathogens are *C. trachomatis* or *N. gonorrhoeae*.
- ✓ Other less common aetiologies include HSV and *T. vaginalis*.

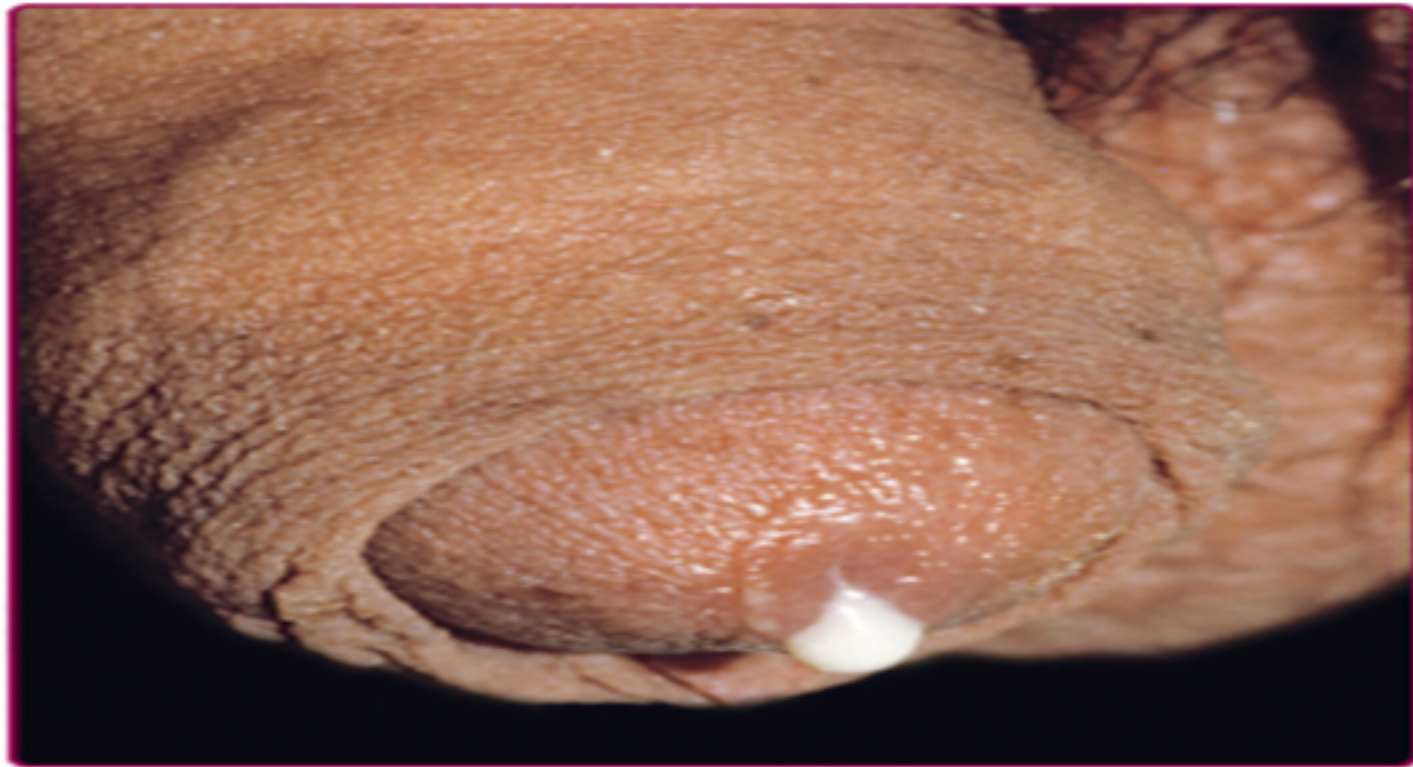
Reproductive Tract Diseases

- Pelvic Inflammatory Disease
 - ✓ Pelvic inflammatory disease (PID) is a polymicrobial infection of upper genital tract structures, namely the uterus, fallopian tubes, and ovaries.
 - ✓ When the endocervical canal barrier is compromised, vaginal bacteria can ascend into the normally sterile space of the upper genital tract (uterus, fallopian tubes, and ovaries).
 - ✓ A sexually transmitted infection affecting the cervix can initiate the processes leading to PID
 - ✓ PID is primarily associated with *N. gonorrhoeae* and *C. trachomatis*, together with enteric gram-negative rods and anaerobes.
 - ✓ Patients can present with a range of symptoms, from lower back pain to fever, chills, lower abdominal pain, and cervical and adnexal tenderness.

Reproductive Tract Diseases

- Urethritis
 - ✓ Urethritis is inflammation of the urethra.
 - ✓ It is usually caused by a sexually transmitted infection, particularly in sexually active men.
 - ✓ Dysuria is a common presenting complaint. Discharge from the urethra, pruritus, and burning are also common complaints.
 - ✓ *Neisseria gonorrhoeae* and *C. trachomatis* are the most common organisms implicated. Other organisms include *Mycoplasma genitalium* and *T. vaginalis*.
 - ✓ Urethral swab for staining, wet mount, Culture & sensitivity, PCR

Reproductive Tract Diseases Urethritis



Thick purulent Urethral Discharge in Gonorrhoea

THANK YOU FOR YOUR
ATTENTION

QUESTIONS